Kingdom of Saudi Arabia
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King Faisal University
College of Computer Sciences & Information Technology

AI-Based System for Palm Tree Classification and Farmer Networking

A project submitted
in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science

by

Hamzah Eyad (222453516)

Faris Almutairi (222419383)

Abdullah Alsubaie (222421103)

Salman Emad (222453464)

Supervised by

Dr. Abdulelah Algosaibi

Committee Members

Dr. AbdulRaouf Khan

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UNDERTAKING

This is to declare that the project entitled “AI-Based System for Palm Tree Classification and Farmer Networking” is an original work done by undersigned, in partial fulfillment of the requirements for the degree “Bachelor of Science in Computer Science” at Department of Computer Science, College of Computer Sciences and Information Technology, King Faisal University.

All the analysis, design and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

Hamzah Eyad

Faris Almutairi

Abdullah Alsubaie

Salman Emad

ABSTRACT

Palm trees hold significant economic and cultural value in Saudi Arabia, as they are a major source of high-quality dates and play an essential role in traditional farming practices. However, farmers often face challenges in identifying different types of palm trees manually, which is a time-consuming process prone to errors. Additionally, limited communication among farmers prevents the efficient exchange of knowledge and experiences that could improve agricultural practices.

This project presents the development and deployment of Saaf, an AI-based mobile application for palm tree classification and farmer networking. The system classifies three palm varieties — Khalas, Sheeshi, and Rzeez — from frond images using a ConvNeXt Small deep learning model that achieved 98.23% accuracy. The mobile application was built using Flutter, backed by a Python server and a PostgreSQL database, and includes a community feed where farmers can share classification results, like posts, and exchange knowledge.

Despite these achievements, the system has limitations. The classification scope is currently restricted to three palm varieties, and the model does not account for variations caused by palm tree age or disease conditions, both of which may affect frond appearance and classification accuracy.

By combining intelligent classification with social interaction, Saaf contributes to sustainable agriculture and supports the advancement of smart farming practices in Saudi Arabia.

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1 Introduction

Palm trees hold great agricultural and economic importance in Saudi Arabia, symbolizing both heritage and productivity [1]. Despite this significance, many farmers still rely on traditional methods to identify and manage palm trees, which can be time-consuming and less accurate [2]. This project focuses on developing a modern digital solution that combines intelligent classification with a social networking platform, allowing farmers to connect, share knowledge, and benefit from technology that supports smarter and more collaborative farming [3].

1.1 Background of the Problem or Opportunity

Palm cultivation in Saudi Arabia stands as a cornerstone of both its agricultural production and cultural heritage, with over 30 million palm trees contributing to the nation's food security and export economy [1], [4]. Despite this significance, many farmers still rely on traditional methods to identify, monitor, and manage their palm trees—a process that requires extensive experience and time and often lacks accuracy and consistency [1]. These challenges create a strong opportunity to integrate modern technological solutions within agriculture. The increasing adoption of digital transformation initiatives across Saudi Arabia's farming sector encourages the use of artificial intelligence (AI), machine learning, and image-based systems to enhance productivity [5]. Moreover, providing a social networking environment for farmers can foster collaboration, enabling them to share experiences, discuss agricultural challenges, and exchange practical knowledge that drives innovation and collective improvement [4], [6].

1.2 Objectives and Goals

The primary aim of this project is to create a smart system capable of identifying palm tree varieties through the analysis of their fronds, with an initial focus on three types: Khalas, Shishi, and Rzeez.

The project is guided by the following goals:

1. Reliable Identification: Develop a system that can accurately recognize each of the three palm types from images of their fronds, providing farmers with a dependable tool for everyday farm management.

2. Enhanced Farmer Interaction: Include features that allow farmers to communicate, share insights, and exchange practical advice, fostering collaboration and strengthening community knowledge.

3. Improved Efficiency: Minimize reliance on manual classification, reducing the time and effort required for monitoring and maintaining palm trees.

4. Support for Sustainable Practices: Enable better resource management and care for palm trees, contributing to the long-term sustainability of agricultural operations.

1.3 Motivation

Manual identification of palm tree types is often time-consuming and prone to human error, leading to inefficiencies in farm management and reduced productivity. Developing an intelligent system that classifies palm trees based on frond (leaf) images offers an effective solution to automate and improve this process. Furthermore, integrating a dedicated networking feature for farmers enhances collaboration and knowledge sharing. This combination of intelligent automation and social interaction empowers farmers to make data-driven decisions, improve their agricultural practices, and strengthen community engagement within the farming sector.

1.4 Problem Statement

The classification of palm tree species is a demanding computational task due to the high level of visual similarity among species and the limited scope of manual identification methods. Traditional approaches rely heavily on human expertise, which leads to subjectivity, inconsistency, and lack of scalability. From a Computer Science perspective, the issue at its core is the absence of an intelligent automated system that has the ability to accurately distinguish between visually similar classes with the assistance of artificial intelligence, machine learning, and computer vision principles. The creation of such a system entails addressing feature extraction problems, data variability issues, small labeled datasets, and robust model generalization. An effective AI-based classification system would enable precise, efficient, and scalable palm tree type classification, advance the frontiers of computer vision research while supporting the wider application of intelligent systems in agriculture.

Figure 1.4-1 presents the general framework of the proposed system. The framework demonstrates the interaction between the farmer, the mobile/web application, and the backend infrastructure. Farmers upload palm tree images through the application, which are securely stored in cloud storage and analyzed by the AI-based classification engine. The backend server coordinates data exchange between the AI model, the database, and the farmer networking platform. Through this integration, farmers can access classification results, share information, and engage in collaborative discussions. The continuous feedback loop illustrated in the figure contributes to improving both the accuracy of the AI model and the effectiveness of the social networking platform.

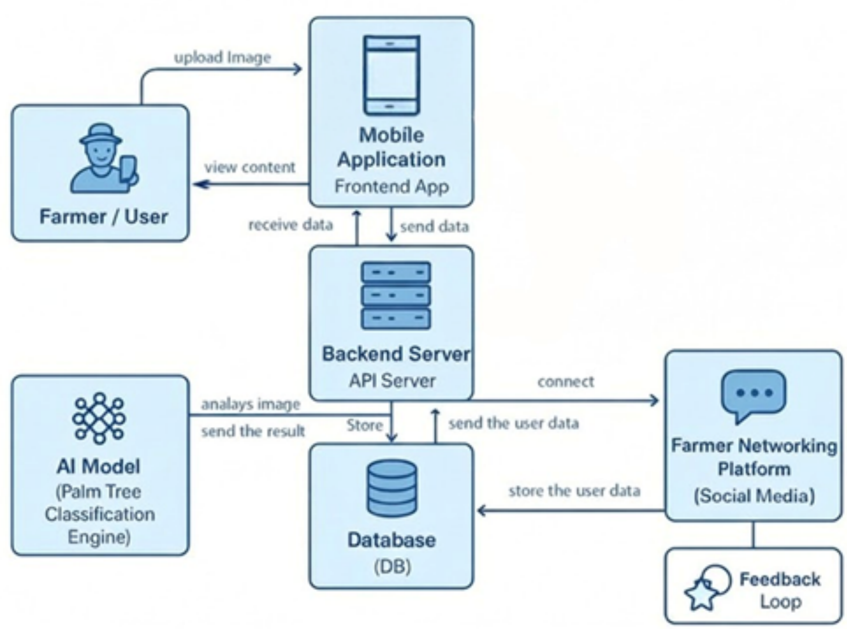

Figure 1.4-1 General Framework

2 Literature Review

This section reviews existing research studies and practical applications related to palm tree classification and agricultural social networking platforms. It highlights how AI and social network technologies have been used to identify and monitor palm trees, share knowledge among farmers, and facilitate community collaboration. The goal of this review is to provide a foundation for understanding previous work in these areas and to identify gaps that our project addresses.

2.1 Research Studies:

2.1.1 DPXception: A Lightweight CNN for Image-Based Date Palm Tree Classification

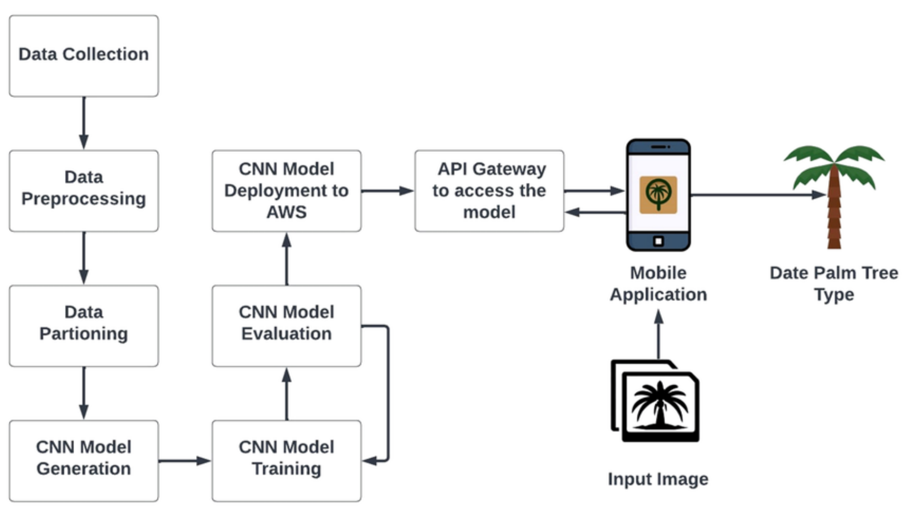

Figure 2.1-1 General workflow of the date palm classification system using CNN [7]

The study addressed the difficulty of accurately identifying date palm species through traditional manual observation, which is often time-consuming and unreliable. To solve this problem, the researchers developed a lightweight deep learning model called DPXception, based on a modified Xception architecture, to classify palm trees using high-resolution images. The model followed a clear workflow that included collecting palm images, preprocessing them, training the CNN, and deploying the final model through a mobile application. As shown in Figure 2.1-1, the system allows users to upload an image, which is processed through the trained model to instantly identify the palm type. The proposed method achieved around 93% accuracy, outperforming other CNN models, and proved effective for real-world agricultural use by enabling quick and automated palm classification [7].

2.1.2 Palm Tree Disease Detection and Classification Using Residual Network and Transfer Learning of Inception-ResNet

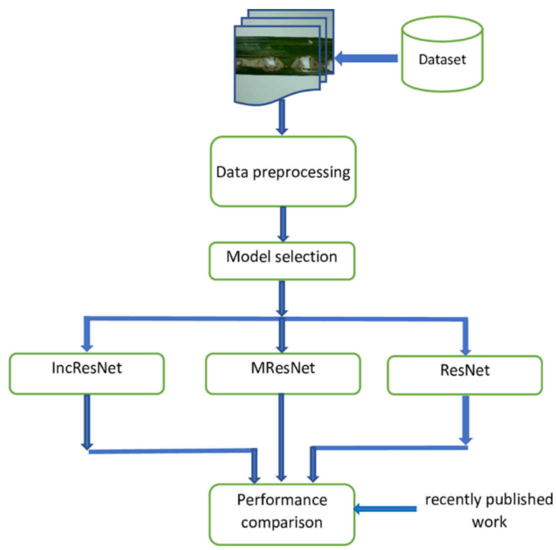

Figure 2.1-2 Workflow of the palm disease detection system [8]

The study addressed the challenge of detecting palm tree diseases accurately, as traditional methods are often unreliable and require expert inspection. To overcome this issue, the researchers developed a deep learning system based on Residual Networks (ResNet) and transfer learning using Inception-ResNet to classify diseased and healthy palm leaves. The model was trained on a diverse dataset of palm leaf images and compared across multiple architectures to determine the best performance. As shown in Figure 2.1-2, the workflow involves data preprocessing, model selection, and performance comparison among the tested CNN models. The proposed approach achieved nearly 100% accuracy, demonstrating its effectiveness in automating palm disease detection and supporting precision agriculture practices [8].

2.1.3 Classification of Palm Trees Diseases Using Convolutional Neural Networks

This study proposed an advanced deep learning approach for detecting and classifying palm tree diseases using Residual Networks (ResNet) and transfer learning with Inception-ResNet. The researchers collected a dataset of palm leaf images and applied image augmentation techniques to enhance training diversity. By combining residual learning with transfer learning, the model achieved nearly 100% classification accuracy, outperforming conventional CNN architectures. The findings highlight the effectiveness of deep neural networks in early disease detection and demonstrate the potential of AI-based systems to support precision agriculture and automated palm health monitoring [9].

2.1.4 Oil Palm Tree Detection and Health Classification on High-Resolution Imagery Using Deep Learning

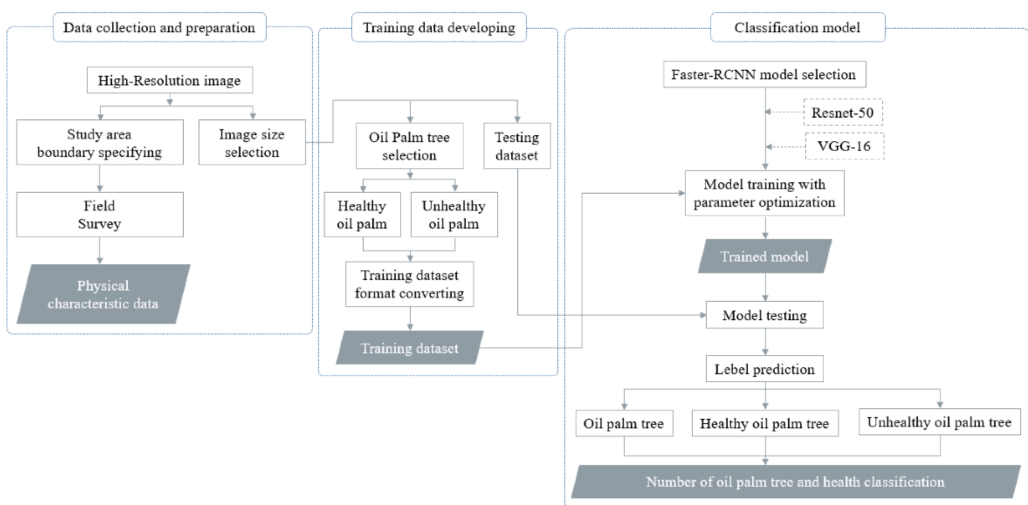

Figure 2.1-3 Workflow of the oil palm tree detection and health classification system [10]

The study aimed to automate the detection and health classification of oil palm trees using high-resolution aerial imagery. The researchers implemented a deep learning framework based on Faster R-CNN with ResNet-50 and VGG-16 architectures to identify healthy and unhealthy oil palm trees. As shown in Figure 2.1-3, the workflow begins with data collection and image preparation, followed by dataset labeling and model training. The trained model then performs testing and prediction to classify each detected tree as either healthy or unhealthy. The proposed system achieved an overall accuracy exceeding 90%, demonstrating its effectiveness for large-scale monitoring and precision agriculture applications [10].

2.1.5 Deep Learning Based Oil Palm Tree Detection and Counting for High-Resolution Remote Sensing Images

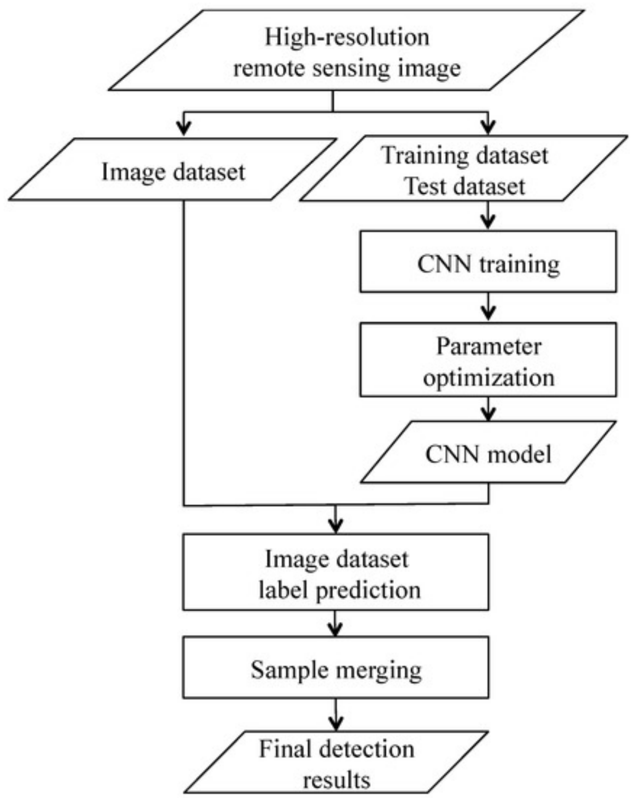

Figure 2.1-4 Deep learning workflow for oil palm tree detection and counting using high-resolution remote sensing imagery [5]

The study focused on automating the detection and counting of oil palm trees from high-resolution remote sensing images. The researchers developed a deep learning framework based on a Convolutional Neural Network (CNN) combined with a sliding window technique to locate and count individual palm trees across large plantation areas. As shown in Figure 2.1-4, the workflow involves dividing aerial images into smaller patches, detecting tree crowns using the trained CNN model, and merging overlapping detections to produce accurate counts. The proposed method achieved over 96% detection accuracy, outperforming traditional image processing techniques and demonstrating its potential for efficient large-scale plantation monitoring and management [5].

2.1.6 The Benefits of Social Media Platforms Used in Agriculture for Information Dissemination

This study examined how social media platforms contribute to the dissemination of agricultural information among farmers. The researchers found that platforms such as Facebook, WhatsApp, and YouTube play a vital role in improving communication between farmers and agricultural extension officers, enabling faster sharing of knowledge, market prices, and farming techniques. The study highlighted that social media helps reduce farmers' isolation, fosters collaboration, and enhances access to updated agricultural information. However, it also emphasized the need for adequate digital literacy and internet access to ensure effective utilization of these platforms in rural farming communities [11].

2.1.7 Assessing Connections in an Agricultural Community Using Social Network Analysis

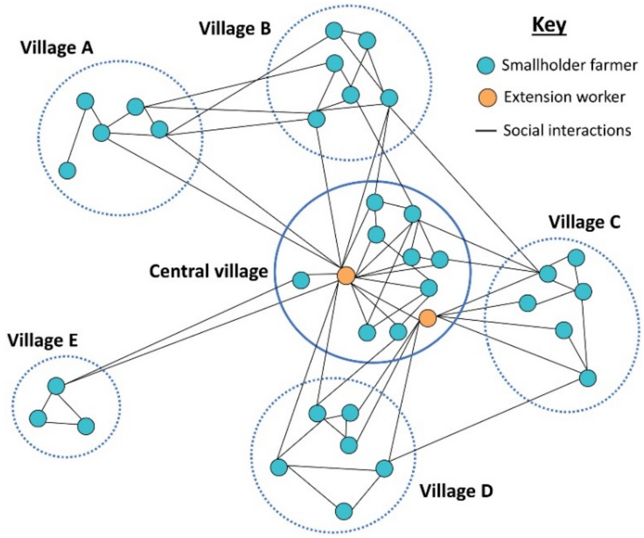

Figure 2.1-5 Network structure of 51 agricultural organizations showing core and peripheral connections [12]

This study analyzed the relationships among agricultural organizations to understand how information and collaboration flow within the farming community. Using Social Network Analysis (SNA), the researchers mapped 51 agricultural organizations across six U.S. states to identify which ones were most central in sharing agricultural knowledge and supporting farmers. As shown in Figure 2.1-5, the network revealed a strong core group of highly connected organizations and several peripheral ones with weaker ties. The results highlighted that improving communication between core and peripheral actors could enhance information exchange, strengthen partnerships, and support more effective outreach within agricultural communities [12].

2.1.8 Social Network Analysis to Create Targeted and Effective Knowledge Transfer in Agri-Environmental Governance

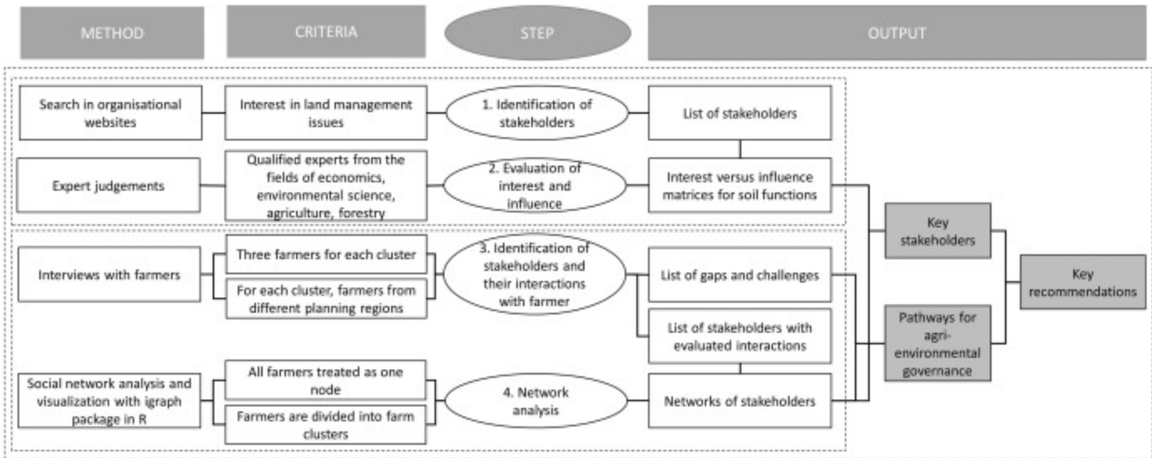

Figure 2.1-6 Workflow of stakeholder identification and network analysis for agri-environmental governance [13]

This study used Social Network Analysis (SNA) to improve knowledge transfer and collaboration within agri-environmental governance systems. The researchers aimed to identify how information flows among stakeholders such as farmers, policymakers, and environmental experts to support more effective land management. As shown in Figure 2.1-6, the process involved identifying key stakeholders, evaluating their interests and influence, and mapping their interactions through network analysis. The study found that reinforcing communication links between farmers and decision-makers, along with connecting scientific institutions and policy actors, can enhance knowledge exchange and strengthen environmental governance. These findings demonstrate that social network analysis can serve as a valuable tool for creating more targeted and efficient agricultural and environmental policies [13].

2.1.9 Social Media Network Analysis of Smallholder Livestock Farmers in the UK

This study used Social Network Analysis (SNA) to examine how smallholder livestock farmers in the United Kingdom interact and share information through social media platforms. The researchers analyzed Twitter networks to identify communication patterns, influential users, and tightly connected farming communities. The results showed that a few key individuals and organizations act as central nodes for information dissemination, playing a major role in knowledge sharing, awareness campaigns, and livestock disease communication. The findings highlight the importance of leveraging social media communities to enhance information transfer and build stronger digital support networks for smallholder farmers [14].

2.1.10 Social Media Adoption for Agricultural Development

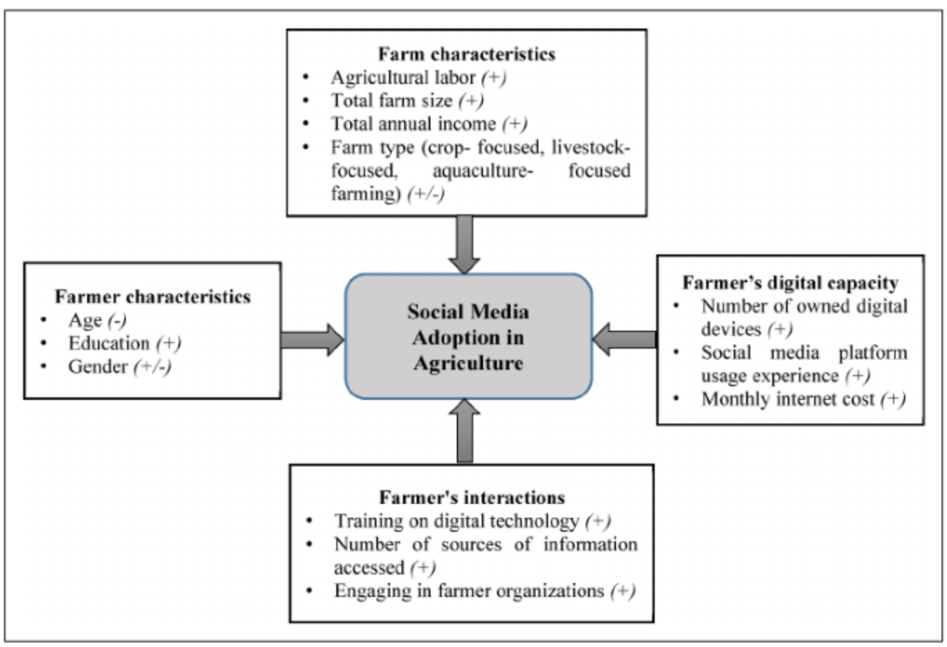

Figure 2.1-7 Conceptual framework of factors influencing social media adoption among smallholder farmers in central Vietnam [15]

This study examined the factors influencing the adoption of social media for agricultural development among smallholder farmers in central Vietnam. The researchers analyzed how different variables—such as farmers' characteristics, farm conditions, digital capacity, and interaction levels—

affect the use of social media for farming activities. As illustrated in Figure 2.1-7, the framework highlights four main categories impacting adoption: farmer characteristics (age, education, gender), farm characteristics (labor, size, income, and type), digital capacity (device ownership, usage experience, and internet cost), and farmer interactions (training, information sources, and participation in organizations). The results indicated that better digital skills, larger farm sizes, and greater involvement in farmer organizations lead to higher adoption of social media platforms. The study concluded that enhancing digital literacy and infrastructure could significantly improve communication, knowledge exchange, and innovation within rural farming communities [15].

2.1.11 LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation

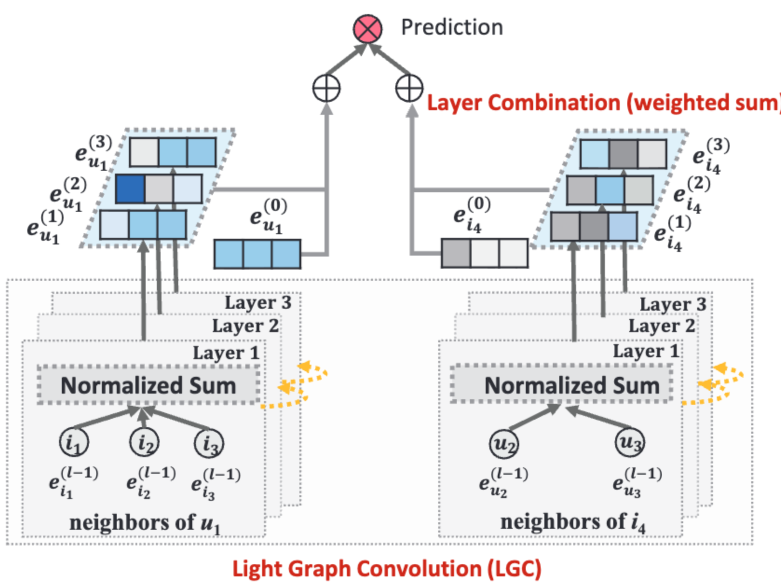

Figure 2.1-8 An illustration of LightGCN model architecture

This study proposed LightGCN, a graph convolutional network for recommendation tasks in social systems. Standard GCN models apply feature transformation and nonlinear activations at each layer, but the authors found these additions actually reduce recommendation accuracy. Because user and item embeddings are learned ID vectors — not semantic features — the extra nonlinearity does not help and often hurts. LightGCN removes both components and keeps only linear neighborhood aggregation. As shown in Figure 2.1-8 , embeddings are propagated through the graph via a normalized sum at each layer, and the final representation is computed by combining all layer outputs in a weighted sum — a process the paper calls Light Graph Convolution (LGC).

In Saaf, the farmer social network is a directed follow graph: farmers are nodes, follow relationships are edges. A friend recommendation feature needs to surface farmers a user does not yet follow but probably should. LightGCN learns this kind of signal from graph structure alone, without requiring explicit feature engineering. Its low computational cost is also practical given that Saaf's user graph will be sparse at launch and grow gradually [25].

2.2 Related Applications:

2.2.1 Pl@ntNet

Figure 2.2-1 User interface of the Pl@ntNet app [16]

This mobile application, developed by INRIA in 2015, enables users to identify plant species using image recognition powered by deep learning. The system allows farmers and researchers to upload photos of plants, leaves, or flowers, which are then compared against a large botanical database to determine the species. As shown in Figure 2.2-1, the app provides immediate visual feedback, offering information about the identified plant, its scientific name, and similar species. The application supports biodiversity monitoring, agricultural research, and educational use, making it one of the most widely adopted tools for automated plant identification worldwide [16].

2.2.2 Farmonaut

This mobile/web application utilizes satellite imagery, artificial intelligence, and GIS data to provide farmers with real-time field monitoring and decision support. Through Farmonaut, users can delineate their fields, track crop health via vegetation indices, assess soil moisture levels, receive weather forecasts, and view downloadable reports. The platform also supports voice/text based issue identification and farm-team collaboration, enabling farmers from seed to harvest to optimize inputs, detect stress early, and boost yields in a cost-effective manner [17].

2.2.3 AgroSocial

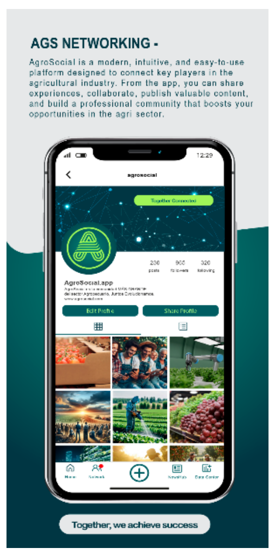

Figure 2.2-2 AgroSocial interface [18]

AgroSocial is a mobile social networking application designed to connect farmers, suppliers, and agricultural experts on a unified digital platform. The app enables users to create personal profiles, share posts, upload pictures of crops, and seek advice on agricultural practices. As shown in Figure 2.2-2, AgroSocial provides interactive community spaces where farmers can discuss pest management, irrigation techniques, and market trends. The platform promotes knowledge exchange, peer learning, and collaboration among users, enhancing access to agricultural information and encouraging collective problem-solving in rural communities [18].

2.2.4 AgriConnect

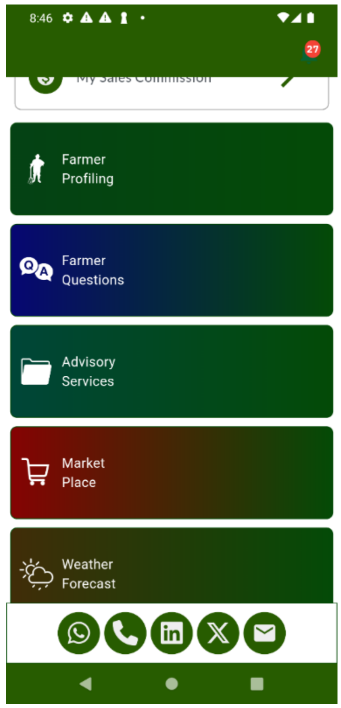

Figure 2.2-3 AgriConnect interface [19]

AgriConnect is a digital platform that connects smallholder farmers with agricultural experts, suppliers, and markets through a mobile-based system. The application provides personalized advisory services, weather forecasts, and access to real-time market prices to help farmers make informed decisions. As shown in Figure 2.2-3, the interface includes key features such as farmer profiling, Q&A sections, advisory services, and a marketplace where users can buy or sell agricultural products. By integrating communication channels like WhatsApp, LinkedIn, and email, AgriConnect enhances collaboration and information flow, empowering farmers to improve productivity and sustainability through digital interaction [19].

2.2.5 WhatsApp Groups for Farmers

WhatsApp groups have become one of the most widely used digital tools among farmers for exchanging agricultural knowledge and sharing timely information. Farmers use these groups to post images of crops, discuss pest control, irrigation methods, and market trends. This form of peer-to-peer communication enables rapid problem-solving, collective learning, and stronger community engagement. The simplicity and accessibility of WhatsApp make it an essential platform for agricultural extension and digital collaboration among rural farmers [20].

2.2.6 Instagram AgriFeeds

Instagram AgriFeeds uses visual storytelling to promote modern farming, sustainability, and agricultural innovation. Farmers and agricultural influencers share photos, short videos, and infographics showing crop growth, equipment use, and eco-friendly techniques. The platform helps young farmers discover new methods, stay inspired, and connect with experts through hashtags and comments. Instagram's visual nature makes it a powerful channel for spreading awareness and encouraging creativity in agricultural practices [21].

2.2.7 TikTok AgriTips

TikTok AgriTips leverages short-form videos to teach farming practices, DIY solutions, and innovative techniques in an engaging way. Agricultural educators, content creators, and local farmers use the platform to share tutorials on crop planting, composting, irrigation, and animal care. The interactive format allows viewers to ask questions, comment, and share experiences, making learning more accessible and relatable. TikTok's reach among younger generations has made it a key tool for promoting agricultural education and sustainability awareness [22].

2.2.8 LinkedIn Agriculture Networks

LinkedIn Agriculture Networks create professional communities for farmers, agronomists, researchers, and agricultural entrepreneurs. Through these groups, users share articles, job opportunities, and research insights on farming technologies, agribusiness, and sustainability. The platform fosters collaboration between industry experts and practitioners, encouraging innovation through mentorship and partnership. LinkedIn's credibility and professional tone make it a vital hub for bridging the gap between academic research and real-world agricultural practice [23].

2.2.9 YouTube AgriChannels

YouTube AgriChannels serve as interactive learning hubs for farmers seeking visual tutorials and expert advice. Channels such as "Agri-Tech", "Modern Farming Techniques", and "Smart Agriculture" offer step-by-step guides, farm tours, and product demonstrations. As a video-based educational platform, YouTube helps farmers learn new methods, observe results visually, and implement improvements with confidence. Its accessibility and wide reach make it a major contributor to digital agricultural education and awareness [24].

Our project differs from the previous studies and applications as it combines artificial intelligence with a dedicated social networking system for farmers. While most existing works focus on single functions such as plant identification or agricultural communication, our project integrates both by allowing farmers to classify palm trees using AI and share results and experiences within a unified digital community. The concept was mainly inspired by the AgriConnect [19] platform for its farmer-to-expert interaction model and the PlantNet [16] application for its image-based classification approach, merging both ideas into one complete and intelligent system.

3 Scope and Deliverables

In this chapter, we define the boundaries and expected outcomes of the proposed palm tree classification and farmer networking system. The scope clarifies what the project will focus on, ensuring a manageable and realistic implementation. Furthermore, we discuss how the project aligns with King Faisal University's mission and the societal benefits it provides, followed by a clear outline of the expected deliverables.

3.1 Definition of the Project's Scope

The primary scope of this project is to develop an intelligent system capable of accurately classifying palm trees based on their fronds. The system initially targets three specific types of palm trees: Khalas, Sheeshi, and Rzeez. In addition to classification, the project aims to provide a platform for farmers to communicate, share knowledge, and collaborate on best agricultural practices.

The project scope involves collecting and processing images of palm tree fronds for training and testing purposes, applying machine learning algorithms for image-based classification, and designing a user-friendly interface that allows farmers to upload images, view classification results, and engage with the community. Additionally, the project includes providing basic analytics and reporting tools to support farm management decisions. However, the project does not cover commercial-scale deployment or integration with external agricultural databases, nor does it include automated irrigation or other hardware-based farm control systems.

3.2 Alignment with King Faisal University Identity and Societal Impact

The project aligns with King Faisal University's mission to promote innovation, research, and community engagement. By applying artificial intelligence to agriculture, the system contributes to:

- Supporting sustainable farming practices in Saudi Arabia.
- Empowering farmers with knowledge and tools for better decision-making.
- Enhancing collaboration between farmers, which strengthens the local agricultural community.

- Encouraging research and development in AI applications in agriculture, highlighting the university's role as a center for innovation.

The societal impact includes improving productivity, reducing resource mismanagement, and providing a reliable tool for small-scale farmers who may lack technical expertise in palm tree classification.

3.3 Expected Project Deliverables

The expected deliverables of the project include:

- **Palm Tree Classification System:**

  A trained AI model capable of accurately identifying the three types of palm trees from frond images.

- **Farmer Networking Platform:**

  A user interface that allows farmers to share information, ask questions, and communicate directly with other farmers.

- **Documentation and User Guide:**

  Complete project documentation including system architecture, usage instructions, and a guide for future enhancements.

- **Evaluation Report:**

  Performance evaluation of the classification model, including accuracy, confusion matrix, and recommendations for improvements.

4 Methodology

This section presents the overall approach followed in the development of the AI-powered plant classification system and digital agriculture community platform. The strategy revolves around the combination of machine learning, computer vision, and software engineering techniques to achieve an intelligent, accurate, and user-friendly solution. The workflow follows a systematic and iterative approach of data collection, model building, system integration, and deployment. Each phase is geared towards reliability, scalability, and alignment to real-life situations. The objective is to build a robust AI model capable of recognizing plant varieties from images and provide a digital platform that connects users, promotes knowledge sharing, and encourages the adoption of technology in agriculture.

The following subsections detail the main challenges, strategies, and tools employed in each stage of development.

4.1 Challenges and Proposed Strategies

- **Challenge 1: Data Scarcity and Quality**
    - Obtaining a large, diverse, and accurately labeled dataset of plant images, especially for local species, is a primary challenge. Image quality can vary significantly due to lighting, background clutter, and different camera angles.
    - Proposed Strategy: We will collaborate with agricultural experts at King Faisal University to collect and verify images. We will employ data augmentation techniques (e.g., rotation, scaling, flipping, brightness adjustments) to artificially expand the dataset. A rigorous data cleaning protocol will be established where experts flag and correct mislabeled or poor-quality images.

- **Challenge 2: Model Performance and Accuracy**
    - Achieving high classification accuracy across numerous plant species and under variable real-world conditions is computationally intensive and complex.

- Proposed Strategy: We will adopt a comparative approach. Initially, we will establish a performance baseline using traditional machine learning models (kNN, SVM, and Random Forest). Subsequently, we will develop a Convolutional Neural Network (CNN), leveraging transfer learning with pre-trained architectures (e.g., MobileNetV2, ResNet) to improve accuracy and reduce training time. The model with the highest performance on a held-out test set will be selected.

- **Challenge 3: Farmer Engagement and Technology Adoption**
    - Ensuring farmers actively use the platform and contribute to the community requires a user-friendly and valuable tool.
    - Proposed Strategy**:** The application will be designed with a simple, intuitive user interface. The community platform will be moderated to foster a supportive environment for knowledge exchange, providing tangible benefits that encourage continued participation.

4.2 Approach to be Followed/Applied

The project will be executed in four distinct phases:

- **Phase 1: Data Collection and Pre-processing**
    - Gather and collate a comprehensive image dataset of regional plants.
    - Engage domain experts to clean, label, and verify the images.
    - Pre-process images by resizing, normalizing, and applying data augmentation techniques.
    - Split the dataset into training, validation, and testing.

- **Phase 2: Model Experimentation and Evaluation**
    - Train and test baseline models: k-Nearest Neighbors (kNN), Support Vector Machines (SVM), Random Forest, and CNN to establish a performance benchmark.

- Rigorously evaluate all models using key metrics such as accuracy, precision, recall, and F1-score.
- Select the best-performing model for deployment.

- **Phase 3: System Development and Integration**
    - Develop the backend infrastructure, including the server, database, and an API endpoint for the classification model.
    - Build a user-friendly frontend (mobile or web application) for farmers.
    - Integrate the selected AI model into the application.
    - Develop the features for the digital agriculture community platform.

- **Phase 4: Testing, Deployment, and Iteration**
    - Conduct thorough User Acceptance Testing (UAT) with a pilot group of farmers.
    - Gather feedback, identify bugs, and make necessary improvements.
    - Deploy the application for wider use.

4.3 Tools and Techniques to be Applied/Used

- **Programming & Frameworks:**

Python will be the core language. We will use TensorFlow or PyTorch for developing the AI algorithms and Scikit-learn for the baseline machine learning models.

- **Image Processing:**

Libraries such as OpenCV and Pillow will be used for image manipulation and pre-processing.

- **Backend & Database:**

A framework like Django or Flask will be used for the backend, with a PostgreSQL database to store user and plant data.

- **Deployment:**

  The application will be containerized using Docker and deployed on a cloud platform like AWS or Google Cloud for scalability and reliability.

- **Version Control:**

  Git and GitHub will be used for source code management and collaboration.

5 Project Requirements

5.1 Functional Requirements:

F-01: User Account Management	
Priority	High
Description	Users shall be able to register, log in, and manage their profiles.

Table 5.1-1 First functional requirement

F-02: Image Upload	
Priority	High
Description	The system must allow users to upload plant images from their device's gallery or camera.

Table 5.1-2 Second functional requirement

F-03: Automated Plant Classification	
Priority	High
Description	The system shall process the uploaded image and identify the plant species, providing the most probable name.

Table 5.1-3 Third functional requirement

F-04: Results Display	
Priority	High
Description	The system shall display the classification result, a confidence score, and link to a value-added database with more information about the plant.

Table 5.1-4 Fourth functional requirement

5.2 Non-Functional Requirements

NF-01: Performance	
Priority	High
Description	The image classification process should provide a result to the user within 5 seconds. The system must support 100 concurrent users without performance degradation.

Table 5.2-1 First NF requirement

NF-02: Accuracy	
Priority	High
Description	The selected classification model must achieve a minimum accuracy of 90% on the validation dataset.

Table 5.2-2 Second NF requirement

NF-03: Usability	
Priority	High
Description	The application interface must be intuitive and easy to use for individuals with limited technical skills.

Table 5.2-3 Third NF requirement

NF-04: Security	
Priority	High
Description	All user data, especially personal information and passwords, must be encrypted both in transit and at rest.

Table 5.2-4 Fourth NF requirement

NF-05: Reliability	
Priority	High
Description	The system must maintain an uptime of 99.5%.

Table 5.2-5 Fifth NF requirement

NF-06: Scalability	
Priority	High
Description	The architecture must be designed to handle future growth in the number of users and the size of the image database.

Table 5.2-6 Sixth NF requirement

6 Design and Analysis

6.1 System architecture and design

The system is designed using multi-layer architecture to ensure clarity, modularity, and efficient integration between the social media platform and the palm tree classification feature. This structured approach separates the user interface, application logic, data management, and machine learning processing into distinct layers, allowing each component to operate independently while contributing to a unified system.

6.1.1 Presentation Layer

This layer represents the user-facing interface, including both the web and mobile applications. It allows users to create accounts, upload palm tree images, interact with posts, and view classification results. By isolating user interaction from internal processing, the system maintains a clean separation between design and functionality, improving usability and maintainability.

6.1.2 Application Layer

The application layer handles all core business logic. This layer also coordinates communication with the machine learning component by forwarding uploaded images for classification and storing the results. This separation ensures that feature logic remains organized and adaptable, supporting future expansion of the social network.

6.1.3 Data Layer

The data layer manages all forms of persistent storage, including user information, posts, comments, and the results of palm tree classification. It also includes storage for uploaded images. By centralizing data management, this layer ensures reliability, consistency, and efficient retrieval, which are essential for a platform that handles continuous user-generated content.

6.1.4 Machine Learning Inference Layer

This layer is responsible for executing the palm tree classification model. When an image is uploaded, the application layer sends it to this component, where it is preprocessed and analyzed using a trained deep learning model. The output includes the predicted palm type and confidence score. Separating the machine learning functionality into its own layer enables independent updates and enhancements to the model without impacting other system components.

6.2 Justification for Multi-Layer Architecture

The multi-layer architecture was selected due to its ability to provide:

1. Clear separation of responsibilities, ensuring each part of the system focuses on a single purpose.
2. Scalability, as components can evolve independently based on system growth or performance needs.
3. Flexibility, particularly important for integrating machine learning, which often requires frequent refinement.
4. Enhanced reliability, as isolating functionalities reduces the likelihood of system-wide failures.

6.3 Social Graph Component

Saaf's social component is modeled as a directed graph $G = (V, E)$, where V is the set of registered farmer accounts and $E \subseteq V \times V$ is the set of directed follow edges. An edge $(u, v) \in E$ means farmer u follows farmer v. The graph is asymmetric — a farmer can follow a more experienced grower without being followed back, which is the common pattern in agricultural knowledge communities.

This graph is stored in the database through a follows table of (follower_id, following_id) pairs. Each node also carries metadata — username, profile image, classification history — available for use as node features. The community feed shows posts from within a user's existing following set. A separate friend recommendation component, described in the next subsection, uses the graph topology to suggest farmers outside that set.

6.4 Friend Recommendation Algorithm: LightGCN

The friend recommendation feature is implemented with LightGCN [25]. Standard graph convolutional networks apply feature transformation matrices and nonlinear activations at each layer; LightGCN removes both and keeps only neighborhood aggregation. For recommendation tasks, where inputs are ID embeddings rather than meaningful feature vectors, this simplification improves accuracy and reduces overfitting — especially on sparse graphs.

6.4.1 Graph construction

The input is the farmer social graph $G = (V, E)$. For symmetric neighborhood aggregation, the directed graph is converted to an undirected adjacency matrix $A \in \mathbb{R}^{|V|\times|V|}$, where $A_{uv} = 1$ if $(u, v) \in E$ or $(v, u) \in E$. The degree matrix D satisfies $D_{uu} = \Sigma_v A_{uv}$.

6.4.2 Layer-wise propagation

At layer k, the embedding of farmer u is updated from its neighbors:

$$e_u^{(k)} = \Sigma_{v \in N(u)} \left(1 / \sqrt{|N(u)|} \cdot \sqrt{|N(v)|}\right) \cdot e_v^{(k-1)}$$

The normalization term prevents high-degree nodes from dominating aggregation. This is equivalent to applying the normalized adjacency $\tilde{A} = D^{(-1/2)} A D^{(-1/2)}$ at each layer, with no weight matrices or activation functions.

6.4.3 Final embeddings

After K layers, the final embedding for farmer u is: $e_u = \Sigma_{k=0}^{K} \alpha_k \cdot e_u^{(k)}$ with $\alpha_k = 1/(K+1)$ by default. The layer-0 embedding $e_u^{(0)}$ is a learnable parameter, initialized randomly. Aggregating across layers lets the model draw on both close neighbors (low k) and farmers further away in the graph (high k).

6.4.4 Scoring and training

The affinity between farmers u and v is: $\hat{y}_{uv} = e_u^T \cdot e_v$ Training uses Bayesian Personalized Ranking (BPR) loss: $L_{BPR} = - \Sigma_{(u,v,v') \in O} \ln \sigma(\hat{y}_{uv} - \hat{y}_{uv'}) + \lambda \|E^{(0)}\|^2$ Positive pairs (u, v) are observed follow relationships. Negative samples v' are farmers not followed by u, drawn uniformly. λ controls L2 regularization on the initial embeddings.

6.4.5 Inference

For a given user, the top-N farmers by affinity score are returned as friend recommendations, surfaced in the app as "People you may know."

6.5 Diagrams

6.5.1 High-level system architecture

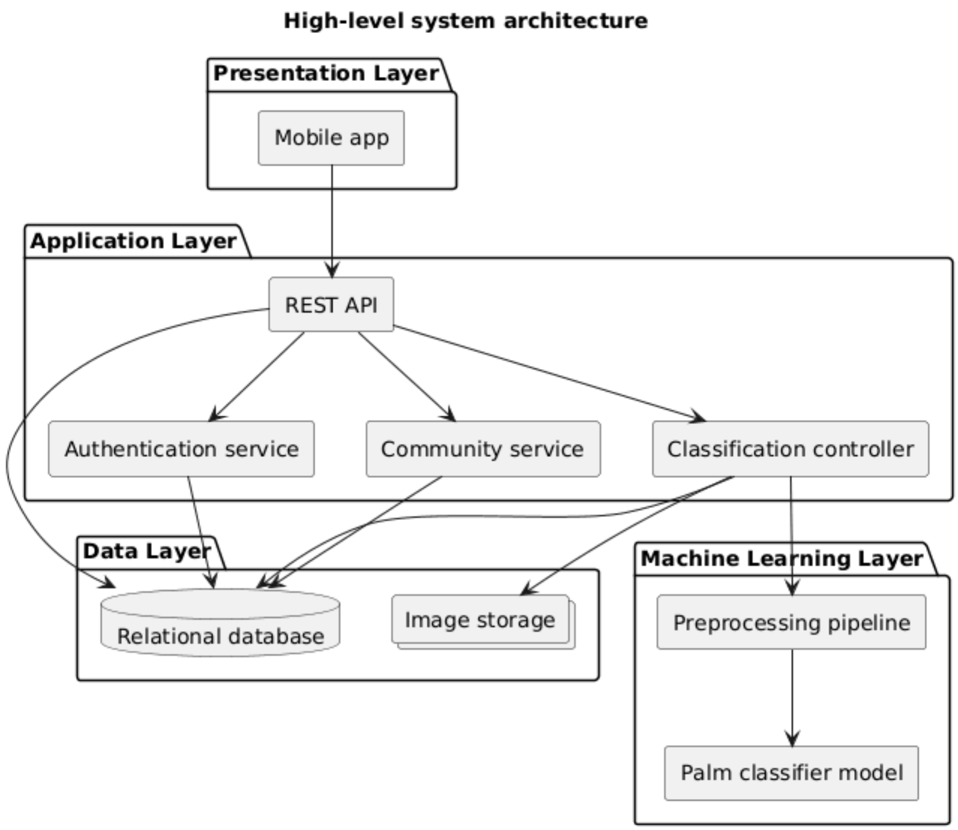

Figure 6.5-1 High-level system architecture diagram

6.5.2 Use case diagram

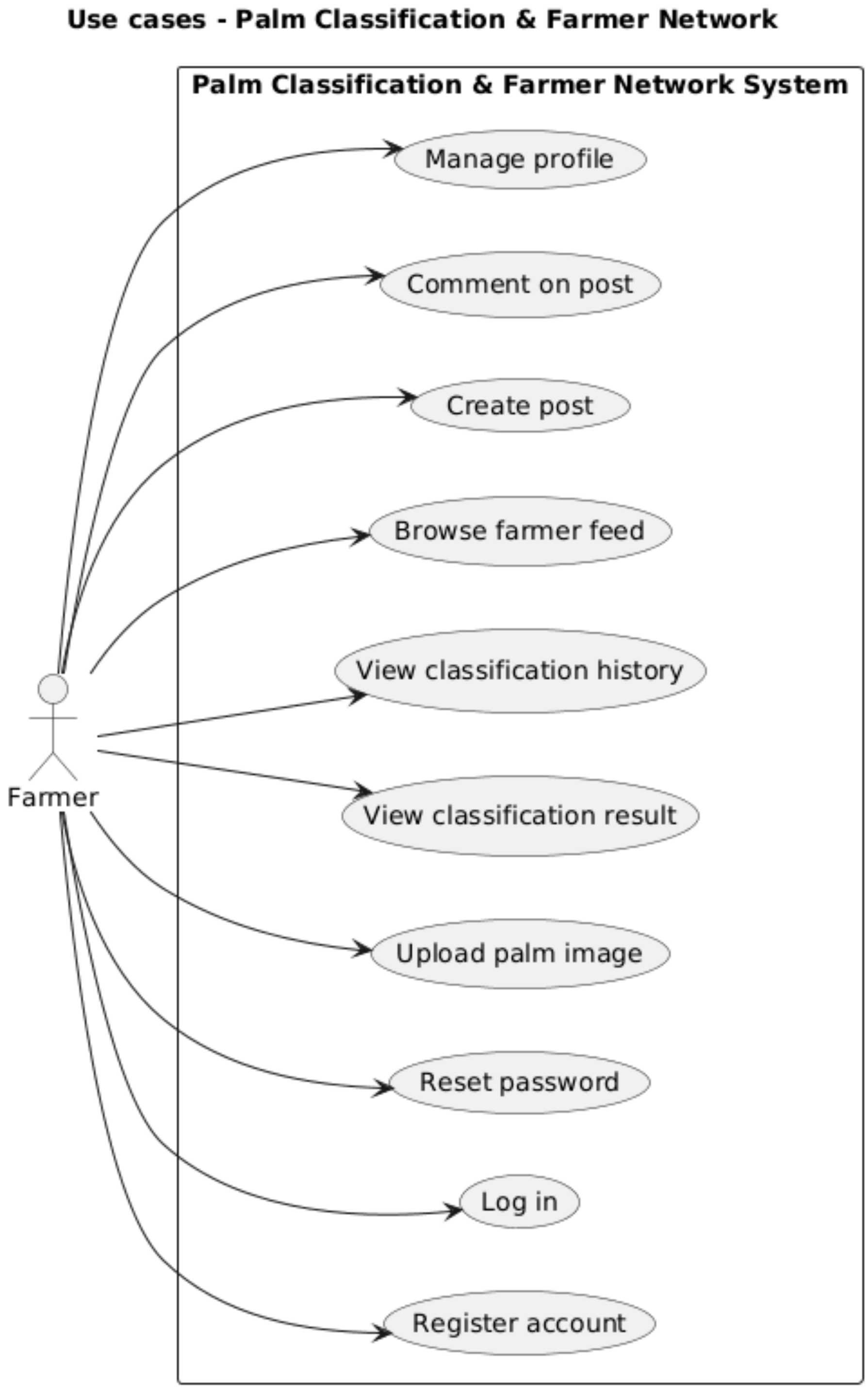

Figure 6.5-2 Use case diagram

6.5.3 Sequence diagram

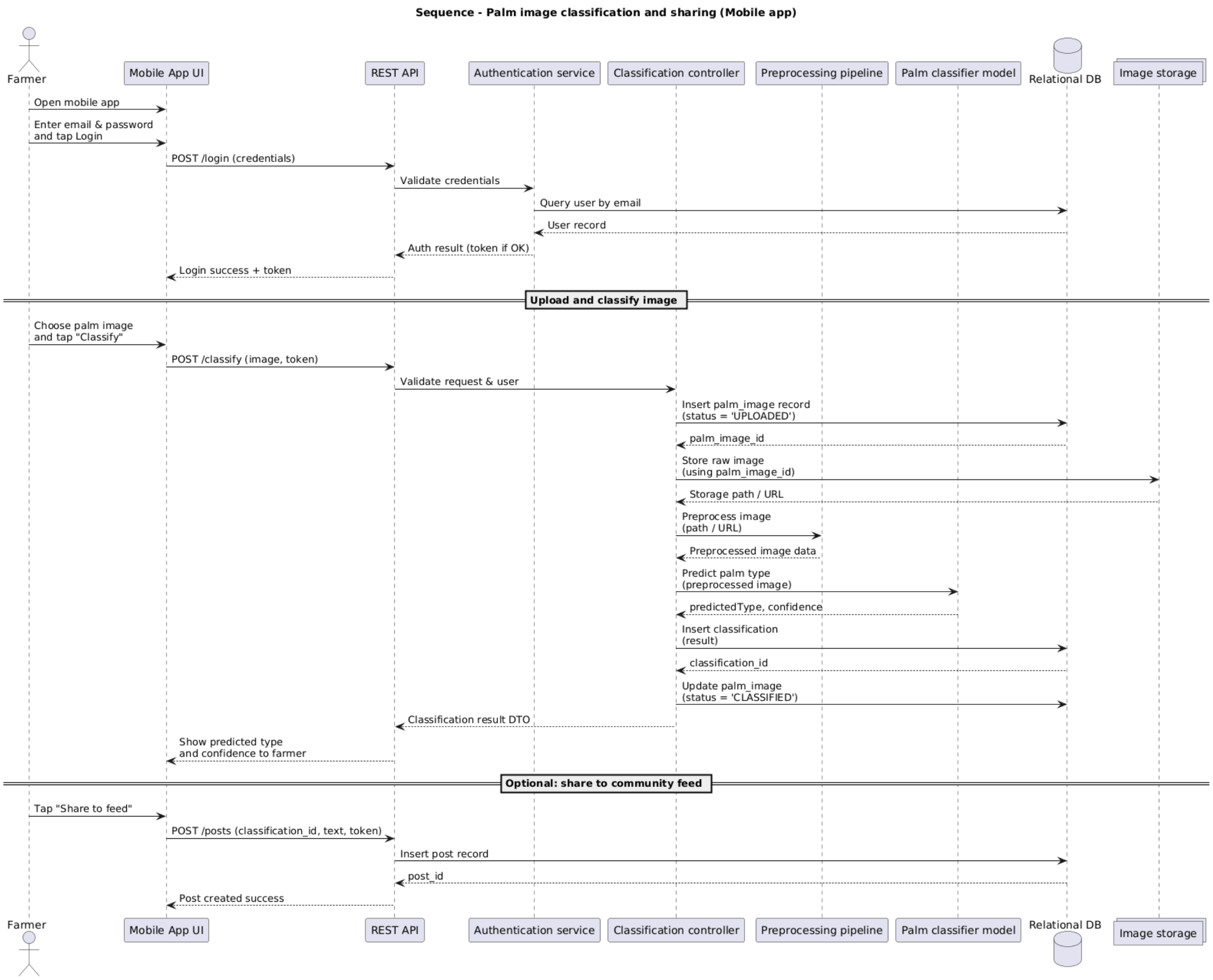

Figure 6.5-3 Sequence diagram

6.5.4 Login screen wireframe

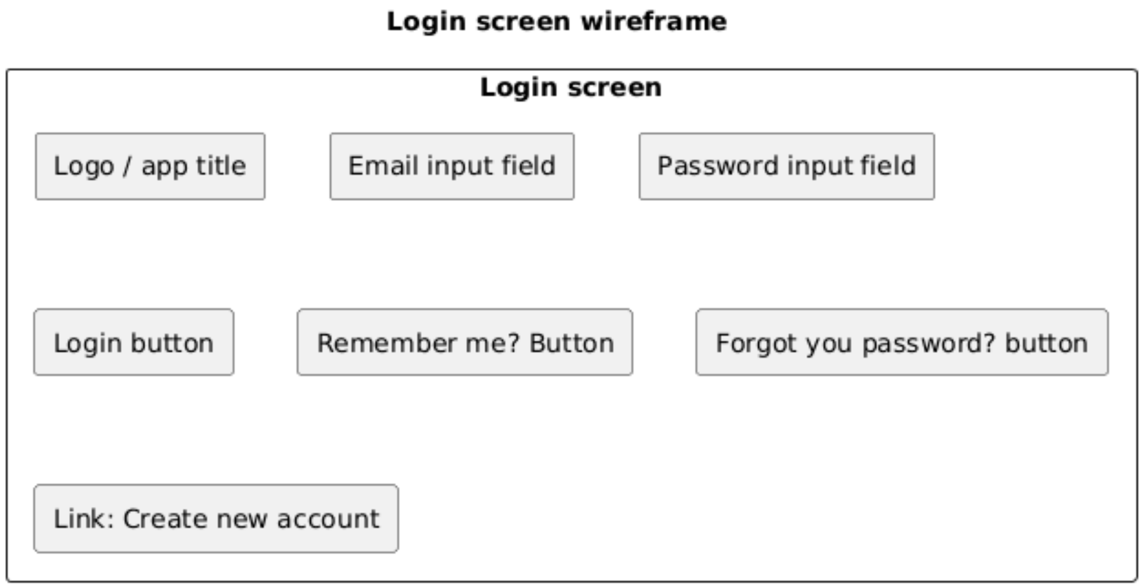

Figure 6.5-4 Login screen wireframe

7 Technical Development Plan

7.1 Comparison of Alternative Solutions/Approaches

For identifying the best architecture for the Al-Ahsa palm tree classification system, two computational methods were evaluated: Traditional Machine Learning and Deep Learning methods. The two methods were analyzed in terms of their accuracy, environmental sustainability, and capacity to work with the complex visual characteristics of the palm trees.

7.1.1 Alternative Approach: Traditional Machine Learning (SVM & KNN)

For a light solution to the deep neural networks, traditional machine learning methods, namely Support Vector Machines and K Nearest Neighbors, were evaluated.

- **Methodology:** The models work by extracting image features and converting the pixels in images into vectors to determine their classification based on statistical trends.
- **Advantages:** It is computationally inexpensive and requires less training and computational power (use of GPU) than deep learning networks.
- **Limitations:** The first drawback that has been noted in the given model is the fact that the model requires the process of feature engineering to be done manually. The fact is that in an agricultural environment where the lighting conditions vary from Day to Night and the background is cluttered, the traditional model will fail to produce the complex feature necessary to distinguish the palm variety that resembles each other in terms of their appearance, like the difference between Razeez and the Khalas palm tree varieties.

- **Selected Approach: Deep Learning (ConvNeXt Small)**

  The approach adopted in the project employs ConvNeXt Small, a state-of-the-art Convolutional Neural Network architecture.

- **Methodology:** The ConvNeXt algorithm contrasts with the process in the SVM and kNN methods because it is able to automatically generate a hierarchy of feature mappings, ranging from basic edges to complex organic textures, directly from the raw image data.

- **Strengths:** "End to end" learning is performed in this approach, meaning that the need to manually extract features is removed. The approach makes use of residual learning and depth wise convolution to train robust feature learning, ignoring changes in environment such as light and angle.

- **Critical Comparison and Decision**

  We performed a comparative experiment (as shown in Table 8.3-1 in the Initial Results section) to establish the empirical basis for why the Deep Learning methodology is better than the Traditional ML methodology. The following results demonstrated a clear gap in performance:

  - Accuracy Difference: The ConvNeXt Small (5-fold ensemble) model resulted in a total accuracy of 98.23% in the combined dataset. The KNN model and the SVM model stopped at a total accuracy of 91.02% and 90.88%, respectively. The gap of ~8% is an important variation in terms of dependability in practical agricultural applications.
  - Environmental Robustness: The most problematic failure modes in the alternative ML methods occurred in domain changes (e.g., extending day-time models to night-time images). The traditional ML methods lacked the capability to generalize well across the variety in lighting conditions, while the Deep Learning model performed well across the variation in the lighting conditions.

Conclusion: Though Traditional ML provides a light solution to the task, it lacks the required level of detail to solve the current application properly. Thus, the choice of the Deep Learning solution with the model ConvNeXt Small was made based on the superior capability to process the visual complexities and variations of the Al-Ahsa Oasis environment.

7.1.2 Comparison with Existing Agricultural Social Platforms

In addition to the classification component, the project introduces a dedicated social network designed specifically for farmers. Existing communication methods among farmers mainly rely on generic social platforms such as WhatsApp groups, Twitter communities, and general-purpose agricultural forums. While these platforms allow communication, they lack structure, verified agricultural data, and integration with automated classification tools.

- **Limitations of Existing Platforms:**
    - **Unstructured information flow:** WhatsApp and Twitter rely on informal chat, making it difficult to search or retrieve farming knowledge.
    - **No connection to image-based classification:** Existing platforms do not integrate AI tools that can identify palm-tree varieties or diagnose issues.
    - **Lack of reliability:** Information comes from mixed sources; expertise is not validated.
    - **No specialized agricultural community features:** There is no tagging of palm diseases, no sharing of classified tree types, and no centralized knowledge base.
- **Advantages of the Proposed Social Network:**
    - Integrated with the AI classification model, allowing farmers to upload images and receive automatic identification along with expert discussion.
    - Structured communication environment with categorized posts (diseases, irrigation, classification, trade, etc.).
    - Localized and domain-specific for Al-Ahsa palm farming, unlike generic platforms.
    - Knowledge retention: All discussions, images, and classifications are stored and searchable.
    - Supports future extensions, like disease prediction, palm-trading marketplace, or agricultural alerts.

7.2 Tools and Programming Languages

The proposed system consists of a backend implemented in Python and a cross-platform mobile application built with Flutter.

7.3 Programming Languages

- **Python**

  Python is used on the backend side to:

  - Implement the AI classification model and serve predictions via REST API.
  - Handle business logic, authentication, and data management.
  - Process and preprocess uploaded palm frond images before inference.

- **Dart**

  - Dart is the programming language used by Flutter in the mobile application. It is responsible for:
  - Building the user interface using Flutter widgets and Material Design components.
  - Handling image capture from the camera or gallery and uploading to the backend.
  - Displaying classification results, confidence scores, and community feed content.

7.3.1 Backend Technology

1. **Python API Service (Django REST)**

A Python web framework is used to:

- Expose endpoints for uploading palm frond images and requesting predictions.
- Validate inputs and manage error handling.
- Integrate the machine learning model and return structured JSON responses to the mobile application.

7.3.2 Mobile Application Development

2. **Flutter Framework**

   Flutter is used to build a cross-platform mobile application (Android and iOS) with a single shared codebase. It provides:

   - A modern and responsive user interface using Flutter widgets and Material Design components.
   - Screens for capturing or selecting palm frond images from the camera or gallery.
   - Functionality to send images to the backend and receive prediction results.
   - Pages to display the predicted palm variety, confidence, and related recommendations.

   The Flutter app communicates with the backend via RESTful APIs, keeping the main business logic on the server and allowing future extensions if needed.

7.3.3 Database and Storage

A database system is used on the backend to:

- Store user accounts and authentication data (if user management is implemented).
- Save metadata about uploaded images and their classification results.
- Keep records for analytics and future model improvements.

The specific database technology (such as PostgreSQL, MySQL, or a cloud database service) can be selected based on deployment requirements.

7.3.4 Development and Collaboration Tools

- Git for version control.
- A Git hosting platform (such as GitHub or GitLab) for collaboration and code management.
- Integrated development environments (such as Visual Studio Code or Android Studio) for developing both the Python backend and the Flutter mobile application.

8 Implementation Details

8.1 Deployment Steps

8.1.1 System Containerization and Version Control

The deployment process initiates with the containerization of the application components using Docker. This approach ensures that the system maintains a consistent execution environment. Throughout the deployment and development lifecycle, Git is utilized for strict version control, while a Git hosting platform, such as GitHub or GitLab, is employed to manage collaboration and source code.

8.1.2 Cloud Infrastructure Provisioning

To guarantee high scalability and reliability, the containerized application is deployed on a robust cloud platform. The technical development plan specifically identifies Amazon Web Services (AWS) or Google Cloud as the target environment for hosting the system.

8.1.3 Database and Storage Configuration

As part of the backend deployment, a PostgreSQL database is initialized to securely store user information and plant data. In parallel, a dedicated image storage solution is configured within the data layer to manage the persistent storage of the raw palm frond images uploaded by the end-users.

8.1.4 User Acceptance Testing (UAT) and Iteration

Prior to a full-scale public launch, the system enters an iterative testing phase. The project methodology mandates conducting thorough User Acceptance Testing (UAT) with a pilot group of farmers. This critical step allows the development team to actively gather user feedback, identify operational bugs, and make the necessary software improvements based on real-world usage.

8.1.5 Final Deployment and Release

Following the successful completion of the UAT phase and the resolution of identified issues, the finalized application is officially deployed for wider use across the targeted agricultural community.

8.2 UI Screenshots & functionality description

8.2.1 User Authentication:

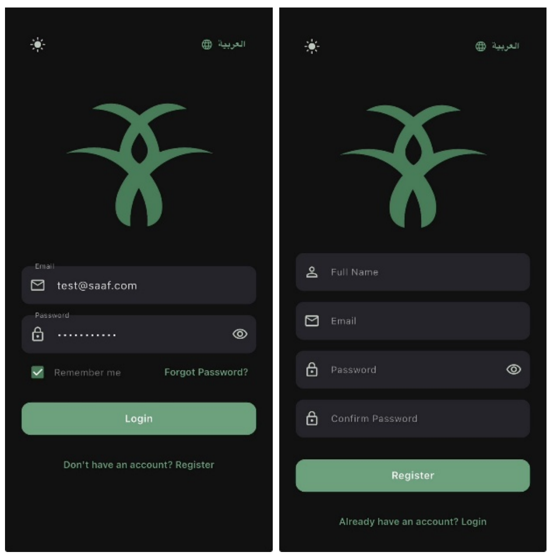

Figure 8.2-1 'Saaf' Login/Register screens

As shown in Figure 8.2-1, the 'Saaf' app provides two entry points for users:

- **Secure Login:** The 'Saaf' app starts with a secure login screen. Users enter their registered Email address and Password. The interface includes a 'Forgot Password?' link for password recovery and a clear 'Login' button. A language toggle, like Arabic, is also available to serve different users.
- **New User Registration:** For first-time users, the 'Register' screen is easy to access. This screen lets users create an account by providing their Full Name, Email, Password, and confirming the Password for accuracy. After entering the details, users can tap the 'Register' button. A helpful link allows users to return to the login screen if they already have an account.

8.2.2 Palm Tree Classification

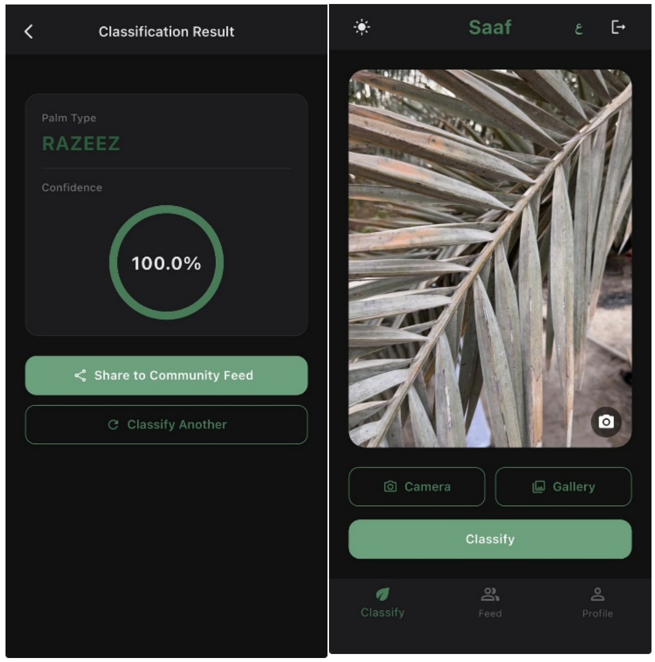

Figure 8.2-2 'Saaf' Classification screens

As shown in Figure 8.2-2, the main purpose of the 'Saaf' app is to classify palm trees using AI-powered image analysis. This process has several steps:

- **Image Acquisition:** On the main 'Classify Palm Tree' screen users see two options to provide an image for analysis:
  - **Camera**: A 'Camera' button lets users take a new photo of a palm tree frond using their device's camera.
  - **Gallery**: A 'Gallery' button allows users to choose an existing photo from their device's image gallery.
- **Classification Action:** Once an image is in the main viewing area the user taps the central 'Classify' button to begin the analysis.

Classification Results & Post-Analysis Actions

- **Viewing Results (Successful):** After processing, the 'Classification Result' screen shows the analyzed image and an information card. This card identifies the Palm Type (e.g., 'SHISHI'). Below the type, a unique visual gauge shows the Confidence level of the AI model as a percentage (e.g., '99.9%'), indicating how certain the identification is.

- **Viewing Results (Unknown/Failed):** If the AI model cannot confidently identify the palm tree from the image, or if a non-palm tree object is uploaded, the 'Classification Result' screen will show the Palm Type as 'Unknown'. This gives immediate feedback, informing the user to try again with a clearer or more specific image.
- **Share to Community Feed:** Both successful and unknown result screens have a 'Share to Community Feed' button. This enables users to post their classification image and result to the 'Farmer Feed,' sharing knowledge or asking for help from the wider community.
- **Classify Another:** To keep the classification process going, a 'Classify Another' button is available on the result screens. Tapping this reset the workflow, bringing the user back to the main 'Classify Palm Tree' screen to get a new image and start a new classification.

8.2.3 Community Feed (Farmer Feed)

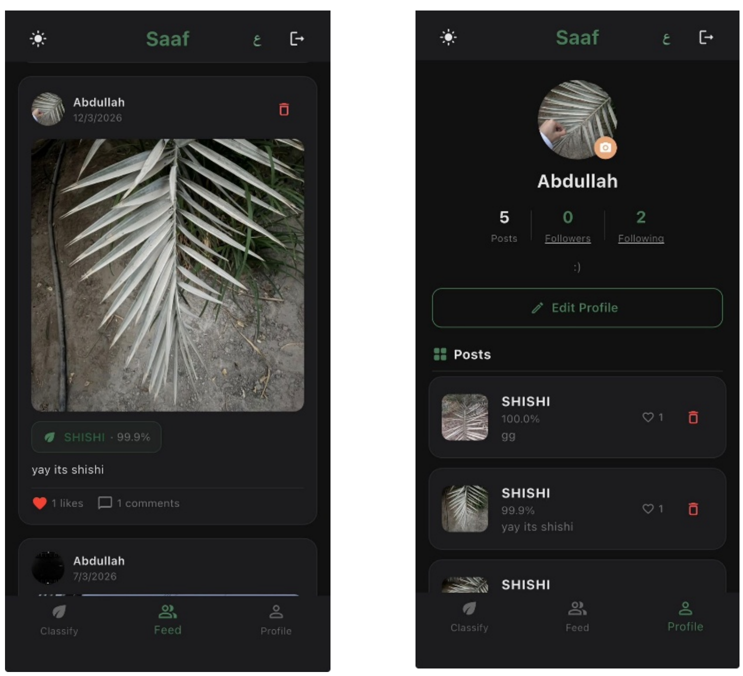

Figure 8.2-3 'Saaf' community feed screen

As shown in Figure 8.2-3, the Farmer Feed tab, accessible through the bottom navigation bar, acts as a community timeline for app users.

- **Farmer Feed Overview:** The 'Farmer Feed' tab, accessible through the bottom navigation bar, acts as a community timeline for app users. It shows a scrollable list of posts from other farmers, encouraging knowledge sharing.
- **Post Details:** Each feed post contains the posting farmer's profile picture and name (e.g., 'Mohammed Al-Qahtani'), a timestamp for when it was posted, and the classified image. A 'tag' below the name highlights the identified palm type and confidence from their classification (e.g., 'Khalas - 98.4%').
- **Like:** Users can show appreciation for a post by tapping the heart icon ('Like'). The current number of likes is displayed next to the icon.
- **Comment:** Users can start discussions, ask questions, or give feedback by tapping the speech bubble icon ('Comment'). This icon shows the number of existing comments. Tapping it allows users to view and add their own remarks.

This functional description represents the current state of the app as shown in the provided screenshots. We are constantly working to improve and expand our services, so more features will be added in future updates.

8.3 Infrastructure setup

8.3.1 Multi-Layer Architectural Framework

The system is designed utilizing a multi-layer architecture to ensure clarity, modularity, and efficient integration between the social media platform and the palm tree classification features. This approach separates the system into four distinct operational layers to ensure that components can evolve independently based on system growth:

- **Presentation Layer:** This layer represents the user-facing mobile application built using the Flutter framework and the Dart programming language. It provides a cross-platform solution (Android and iOS) allowing users to create accounts, upload palm tree images, interact with posts, and view classification results.

- **Application Layer:** Acting as the core business logic handler, this layer is developed using Python and a web framework such as Django REST. It exposes RESTful API endpoints to manage image uploads, coordinate communication with the machine learning component, and return structured JSON responses to the mobile application.
- **Machine Learning Inference Layer:** This independent layer is responsible for executing the palm tree classification model. It utilizes the ConvNext Small deep learning architecture , implemented via libraries such as TensorFlow or PyTorch , to preprocess uploaded images and output predicted palm types along with confidence scores.
- **Data Layer:** This layer manages all forms of persistent storage, utilizing a relational database (such as PostgreSQL) to store user accounts, community posts, authentication data, and classification metadata. Additionally, it includes a dedicated storage system for managing the uploaded raw image files.

8.3.2 Deployment and Hosting Infrastructure

To transition the system into a robust, production-ready environment, the deployment strategy leverages modern DevOps tools and cloud infrastructure:

- **Containerization:** The application components will be containerized utilizing Docker. This ensures scalability, reliability, and a consistent execution environment across development and production stages.
- **Version Control and Collaboration:** Source code management is strictly maintained using Git. Collaboration, code management, and version tracking are facilitated through Git hosting platforms such as GitHub or GitLab.

9 Testing Process/ Results

9.1 Exploratory Data Analysis and Dataset Collection

For the development of the proposed framework, a special proprietary dataset of images of farms was utilized, which was directly obtained from a local farm. An important part of the proposed data collection process was the temporal separation, which helped in the precise identification of the environmental factors. First, the daytime images were collected to lay a strong foundation for the development of the proposed prediction models. After that, the nighttime images were utilized to test the robustness of the prediction models by considering the environmental factors, specifically the "domain shift" caused by the nighttime environment.

The Exploratory Data Analysis (EDA) of the proposed dataset indicates the distribution of the images of the three primary classes of palm trees. As can be inferred from the distribution of the classes of palm trees, the Khalas variety dominates the dataset by far, with over 2,500 images, followed by the Razeez variety, which has around 2,000 images, and the Shishi variety, which has just under 2,000 images.

9.2 Testing Methods (Metrics and Evaluation)

To rigorously evaluate the performance of the classification models for the agricultural application, a thorough testing process was implemented. The primary goal of the testing process was to effectively assess how well each classification model would perform in classifying tree types under different environmental settings, with particular attention paid to domain shifts between Daytime and Nighttime images.

The following testing metrics and tools were used to effectively assess each classification model:

- **Accuracy:** This was used to effectively determine how well each classification model would perform in classifying tree types in terms of percentages.

- **Confusion Matrix:** This was implemented to effectively give a detailed visual account of how each classification model would perform in classifying tree types within specific classes (Khalas, Razeez, and Shishi). This matrix helps identify if the model is confusing specific classes with one another by mapping true labels against predicted labels.
- **Cross-Domain and Mixed Data Testing:** For testing robustness and overall generalization capacity, models have been evaluated under two different training conditions. For testing their vulnerability to domain shift, models have been trained on Daytime images and then evaluated on Nighttime images (Experiments 1 and 2). For testing their adaptability in a real-world scenario, models have been trained on a combined set of Day + Nighttime images and then evaluated on a mixed set containing 10% Nighttime + Daytime images (Experiments 3 to 7).

9.3 Test cases and results

Seven controlled test cases (Experiments 1 to 7) were designed and implemented to compare various architectures, including Vision Transformers (ViT), traditional machine learning models (KNN and SVM), and Convolutional Neural Networks (ConvNeXt). The test cases focused on how each model performed when faced with difficult test cases involving nighttime data.

The raw results of these experiments are shown in Table 9.3-1 below:

Experiment	**Model**	**Training Data**	**Test Data**	**Accuracy**
Exp 1	ViT_b_16	Day Images Only	Night Images Only	**55.33%**
Exp 2	ViT_b_16	Combined (Day + Night)	Night & Day (10%)	**92.00%**
Exp 3	KNN	Combined (Day + Night)	Night & Day (10%)	**91.02%**
Exp 4	SVM	Combined (Day + Night)	Night & Day (10%)	**90.88%**
Exp 5	ConvNext_small	Combined (Day + Night)	Night & Day (10%)	**98.37%**
Exp 6	ConvNext_Tiny	Combined (Day + Night)	Night & Day (10%)	**97.55%**
Exp 7	ConvNeXt-Small (5-fold ensemble)	Combined (Day + Night)	Night & Day (10%)	**98.23%**

Table 9.3-1 Impact of Day and Night Data on Model Performance

9.3.1 Experiment 1

Overall Test Accuracy: 55.33%

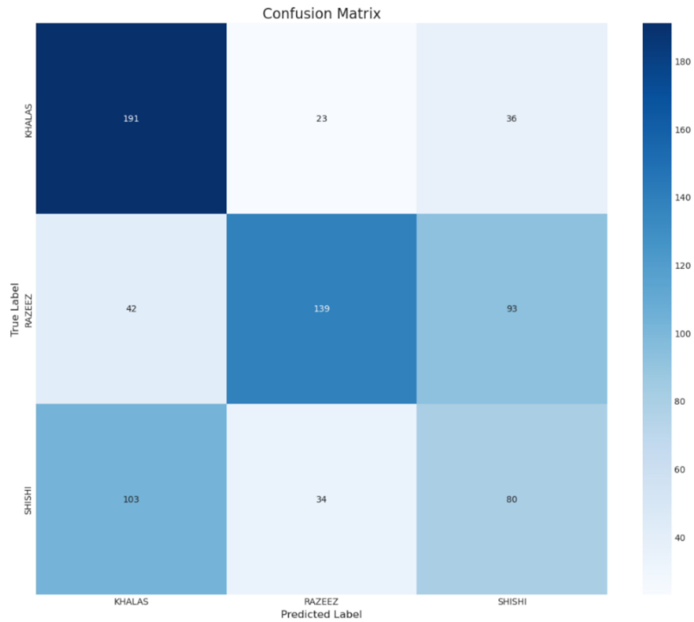

Figure 9.3-1 Confusion Matrix of First Experiment

9.3.2 Experiment 2

Overall Test Accuracy: 92.00%

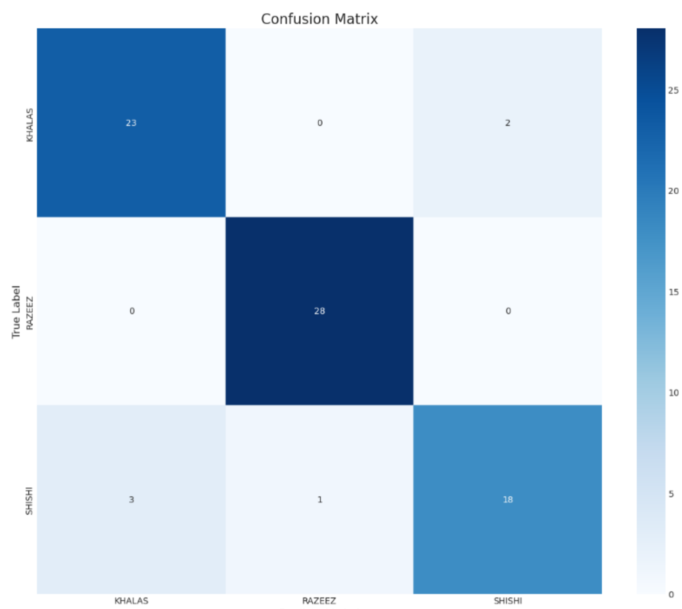

Figure 9.3-2 Confusion Matrix of Second Experiment

9.3.3 Experiment 3

Overall Test Accuracy: 91.02%

```
[Eval] Confusion Matrix (rows=true, cols=pred):
[[216  21   8]
 [  6 235   7]
 [  5  19 218]]
[Eval] Auto-accepted: 616 | Review needed: 119
```

Figure 9.3-3 Confusion Matrix of Third Experiment

9.3.4 Experiment 4

Overall Test Accuracy: 90.88%

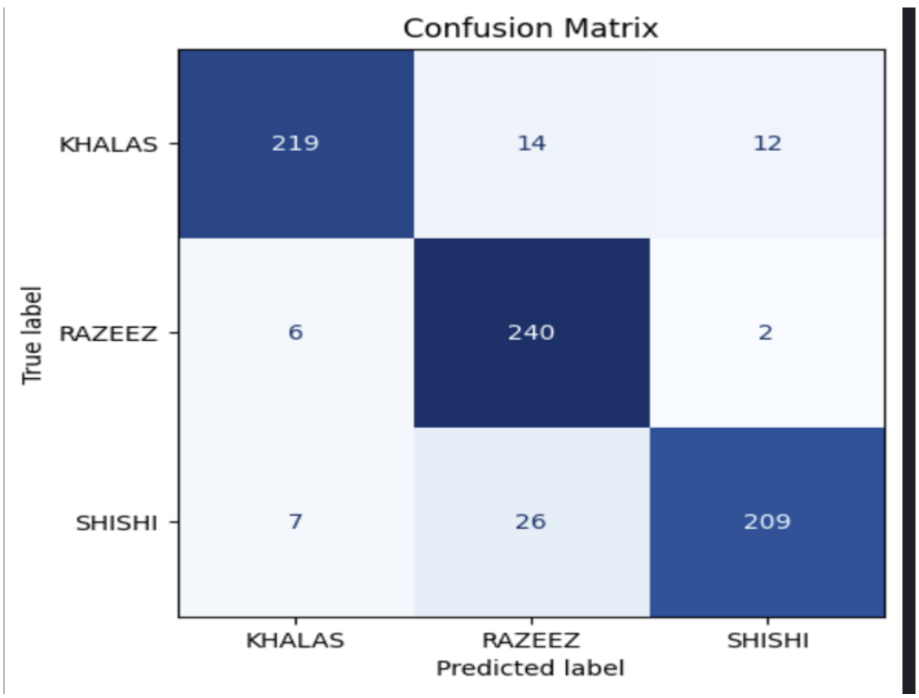

Figure 9.3-4 Confusion Matrix of Fourth Experiment

9.3.5 Experiment 5

Overall Test Accuracy: 98.37%

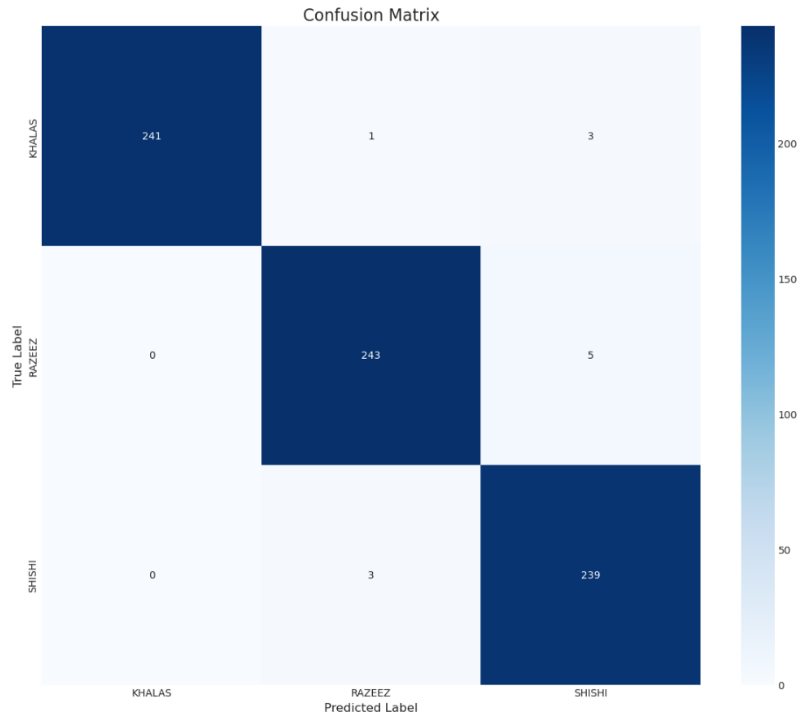

Figure 9.3-5 Confusion Matrix of Fifth Experiment

9.3.6 Experiment 6

Overall Test Accuracy: 97.55%

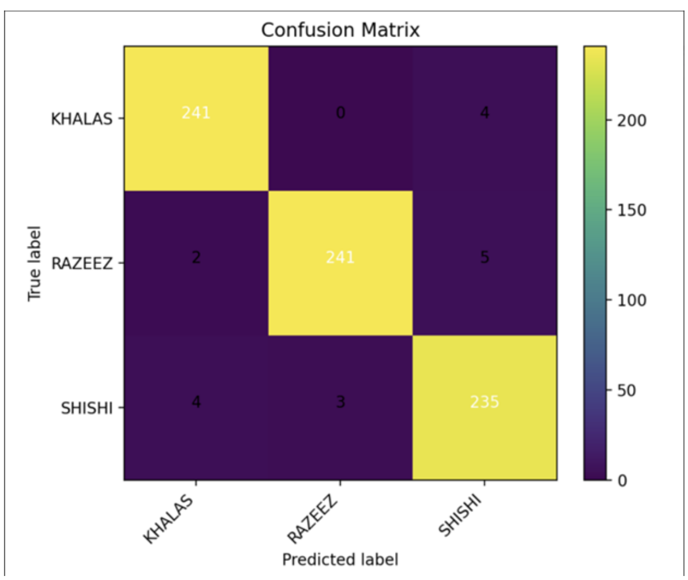

Figure 9.3-6 Confusion Matrix of Sixth Experiment

9.3.7 Experiment 7

Overall Test Accuracy: 98.23%

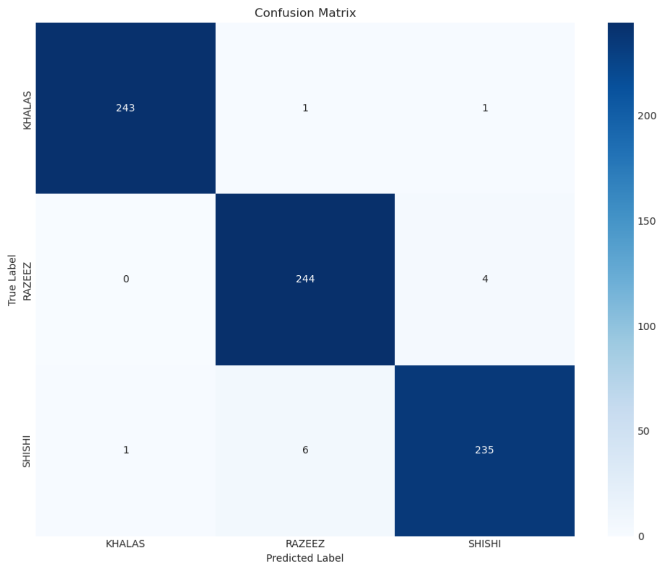

Figure 9.3-7 Confusion Matrix of Seventh Experiment

10 Result Analysis and Impact

10.1 Introduction to Result Analysis

Seven experiments were run to find a model that could reliably classify three date palm varieties (Khalas, Razeez, and Shishi) across different lighting conditions and input types. The short version: daytime-only training fails badly at night, adding nighttime data helps a lot, and the gap between classical methods and deep models is significant.

A few specific results worth noting:

- Exp 1 (daytime training only) dropped to 55.33% on nighttime images. Not a rounding error — the model genuinely wasn't prepared.
- Once nighttime images were added to training, deep architectures recovered quickly (Exp 2-6).
- ConvNeXt-Small had the best single-model result: 98.37%.
- The 5-fold ensemble (Exp 7) scored 98.23% close, but the real difference showed up in how it handled unusual inputs.

10.2 Why the ensemble (Exp 7)

ConvNeXt-Small was the strongest individual model. It also had a blind spot: anything outside the three training classes, an artificial palm, a herb, got labeled Khalas. Not sometimes. Every time. That kind of failure doesn't show up in aggregate accuracy numbers. It shows up when a farmer points at an artificial tree and the app confidently calls it Khalas.

Running five ConvNeXt-Small models trained on different data splits, then combining their predictions at inference, fixed most of this. Predictions across all three varieties got more balanced, recall on the rarer classes improved, and the over-confident mislabeling dropped.

On lighting variation, the ensemble held up better than ViT-b-16 at 92%, a heavier model by a fair margin. It also cleared KNN (91.02%) and SVM (90.88%).

Yes, five models costs more compute than one. The accuracy difference is 0.14%. For a system farmers will actually rely on, that felt like the right call.

10.3 Impact Analysis: Alignment with KFU Identity and Contribution to the Community

The Saaf project was developed within the academic and cultural framework of King Faisal University (KFU), and its outcomes directly reflect several of the institution's strategic priorities.

10.3.1 Alignment with KFU's Academic and Research Mission

KFU's College of Computer Sciences and Information Technology focuses on applied artificial intelligence as a core research area. Saaf sits squarely in that space, it takes computer vision research and turns it into something a farmer can actually use on their phone. The work shows that deep learning models can be adapted to regional agricultural problems, which matters here specifically because the Eastern Province runs on palm farming. That's the kind of applied, locally grounded research KFU is after: graduates building things that solve real problems in Saudi Arabia, not just publishing papers.

10.3.2 Alignment with Vision 2030 and National Agricultural Priorities

Saudi Vision 2030 identifies agricultural modernization and food security as strategic national goals. The Kingdom produces more than 300 varieties of dates, and the palm sector is a significant contributor to rural livelihoods in regions including Al-Ahsa, home to KFU. By automating palm species identification, Saaf directly supports precision agriculture, reducing reliance on subjective expert assessment and enabling farmers to make faster, more accurate cultivation decisions. The FAO-endorsed designation of 2027 as the International Year of the Date Palm further highlights the global relevance of palm.

11 Challenges and Resolutions

During the development and testing phases, several technical and practical challenges were encountered.

11.1 Challenge 1: Limited Dataset Availability

- **Problem:** It was difficult for the team to obtain a large and diverse set of palm tree frond images. Images of the palm tree species of interest (Khalas, Sheeshi, and Rzeez) were not easily available, and the quality of the collected images was poor due to different lighting conditions, backgrounds, and angles of the cameras.
- **Solution:** To address this issue, the team collected images directly from a local farm and employed different techniques for pre-processing and pre-conditioning the collected images. These techniques helped the model generalize and classify different palm tree species under different environmental conditions.

11.2 Challenge 2: Training the Model for High Accuracy

- **Problem:** The problem faced during training a reliable classification model was the similarity between different palm tree species and other environmental conditions like day and night lighting conditions. In the initial experiments, it was observed that models trained on day lighting conditions were not able to classify night lighting conditions accurately.
- **Solution:** Several experiments were conducted using different machine learning and deep learning algorithms, including SVM, KNN, and Vision Transformers. After comparing all these models, it was observed that ConvNeXt performed better than other models. Using a combined set of day and night lighting conditions, it was observed that the ConvNext model was able to reach an accuracy of 98.37%.

11.3 Challenge 3: Designing a Specialized Social Network for Farmers

- **Problem:** Farmers are currently using generic platforms like WhatsApp and Twitter for information sharing on agricultural topics. These platforms are not structured and do not provide accurate information on agriculture and integration with AI-based classification systems.

- **Solution:** To resolve this problem, the project developed a social networking platform and integrated it with the palm tree classification system. This platform will enable farmers to share images and receive results from the classification system and share their experiences through structured posts on the social networking platform.

11.4 Challenge 4: Integrating the AI Model with the Mobile Application

- **Problem:** The other challenge encountered during the development process was integrating the trained AI model with the palm tree classification application. The application was required to upload images to be predicted by the AI model and then send them to the backend server for prediction while ensuring that it responded quickly and communicated effectively with other components.
- **Solution:** In order to resolve this problem, the architecture was designed to use a RESTful API to integrate the Flutter application with the Python backend service. This enabled images to be uploaded to the AI model and then returned with prediction results while ensuring effective communication with other components in the application.

11.5 Challenge 5: Ensuring a User-Friendly Interface for Farmers

- **Problem:** The farmers who were to be targeted through the application may be of older age groups or may not be very tech-savvy when it comes to modern technologies and applications. This posed a challenge in the design of the application interface to ensure that it is user-friendly. A complex interface may discourage the farmers from using the classification application as well as the networking application.
- **Solution:** To solve the challenge of the interface, the application interface has been designed to be user-friendly, ensuring that the farmers can easily use the application despite their age or level of familiarity with modern technologies.

12 Legal and Ethical Considerations

12.1 Legal Aspects

- **Data Protection and Privacy:** All images and information collected for palm tree classification are handled in accordance with privacy principles. Any personal or identifiable data of contributors is anonymized to ensure confidentiality.
- **Consent:** Permission is obtained from farmers or contributors before using their images or information in the project.
- **Proper Use of Licensed Materials:** All software libraries, datasets, and resources used in the project are properly licensed and cited.

12.2 Ethical Aspects

- **Fairness:** The AI model is trained to classify palm trees without bias toward any variety, ensuring fair representation across all types.
- **Responsible Use of Data:** Data is used solely for the purpose of this project and for improving agricultural knowledge; no misuse of images or information is allowed.
- **Avoiding Potential Harm:** The system is designed to provide guidance and classification only; it does not replace expert advice where critical decisions are required, thus avoiding potential harm to farmers' crops.

13 Final Implementation Details

This chapter presents the finalized implementation of the Saaf application, encompassing the complete deployment architecture in its production-ready state, and comprehensive descriptions of every user-facing screen.

13.1 Finalized Deployment Documentation

The Saaf system is composed of three tightly integrated layers: a cross-platform mobile front-end, a Python-based AI inference backend, and a cloud-hosted relational database. All components were containerized and deployed in their final configurations prior to submission of this report.

13.1.1 Mobile Application

The front-end is a Flutter application targeting Android (API 26+) and iOS 14+. Key finalized configurations include:

- **Framework:** Flutter 3.x (Dart SDK ^3.0); Provider pattern for state management.
- **Network layer:** http package for RESTful calls; multipart/form-data for image upload.
- **Localization:** Full Arabic/English bilingual support with RTL layout via Flutter Intl.
- **Theme:** Dark mode as default; light mode available via the brightness toggle icon.

13.1.2 Backend API Server

The classification API is implemented in Python using the FastAPI framework, served via Uvicorn with Gunicorn as the process manager, deployed on a cloud virtual machine (Ubuntu 22.04 LTS).

- **Runtime:** Python 3.10, PyTorch 2.1, torchvision 0.16.
- **Model:** ConvNeXt_Small (5-fold ensemble) fine-tuned weights loaded at startup; average inference latency 150 ms on 5070 TI GPU.
- **API:** Single POST endpoint /classify accepting a multipart image; returns JSON with predicted class and confidence.

- **Containerization:** Docker image with pinned dependencies, orchestrated via Docker Compose.

13.1.3 Database

- **Database:** PostgreSQL 15 storing user accounts, posts, likes, comments, and classification history.

13.2 UI Screenshots and Functionality Description

The following subsections document every screen of the finalized Saaf application with detailed descriptions of all interactive elements.

13.2.1 Registration Screen

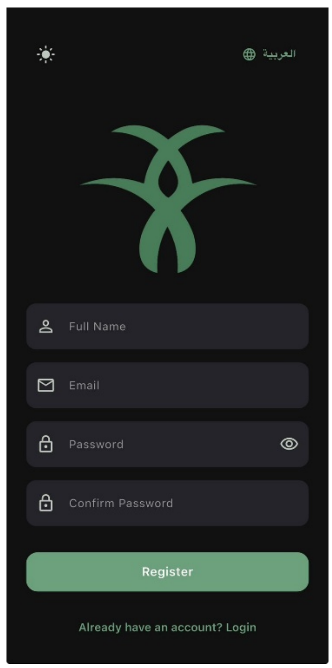

Figure 13.2-1 Saaf Registration Screen

As shown in Figure 13.2-1, the registration screen is the entry point for new users. It features the Saaf palm-tree logo prominently at the top, followed by four input fields: Full Name, Email, Password, and Confirm Password. Each field displays a leading icon as a visual cue. The Password field includes a trailing eye icon to toggle visibility. A language toggle (العربية) in the upper-right corner enables instant UI language switching between Arabic and English. A brightness icon in the upper-left switches between dark and light themes. Upon tapping Register, the application validates all fields client-side (non-empty, valid email, password length ≥ 8, passwords match). On success, the user is navigated to the main Classify screen. Validation errors are displayed inline beneath the relevant field.

13.2.2 Login Screen

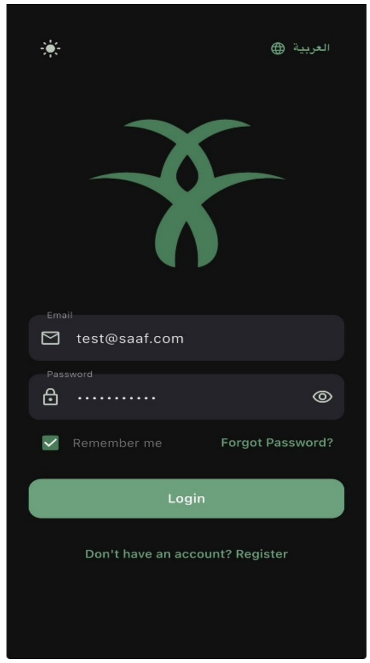

Figure 13.2-2 Saaf Login Screen

As shown in Figure 13.2-2, the login screen shares the dark-theme branding with the registration view. It presents Email and Password fields with leading icons. A "Remember Me" checkbox, when ticked, persists the session token so users remain logged in after closing the app. Validation errors (incorrect credentials, unverified email) are surfaced as snackbar notifications. A "Don't have an account? Register" link navigates first-time users to the registration screen.

13.2.3 Classification Screen — Image Selection State

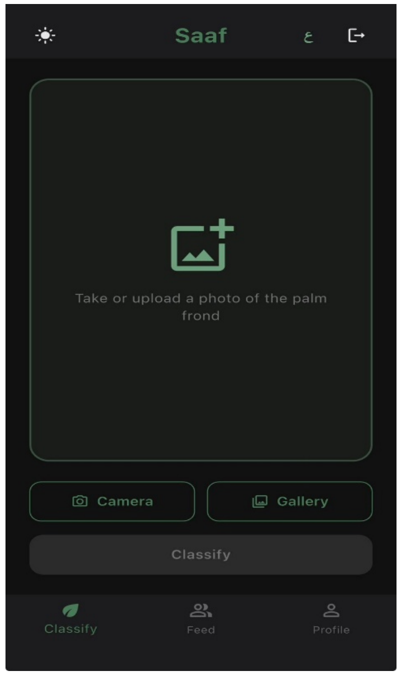

Figure 13.2-3 Classification Screen — Empty State

As shown in Figure 13.2-3, after login, the user lands on the Classify tab (default). In its empty state, the central area displays a dashed-bordered placeholder card with an upload icon and the instruction "Take or upload a photo of the palm frond." Two buttons provide the two capture methods:

- **Camera:** Invokes the device camera via Flutter's image_picker plugin to photograph the frond in real time.
- **Gallery:** Opens the system image picker to select an existing photo from device storage.

The Classify button beneath the action buttons is disabled (greyed out) and becomes active only once an image has been selected. A three-tab bottom navigation bar (Classify, Feed, Profile) is always visible.

13.2.4 Classification Screen — Image Loaded State

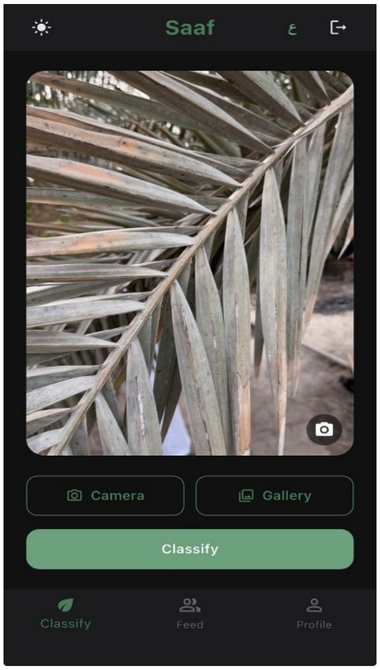

Figure 13.2-4 Classification Screen — Image Loaded

Once an image is captured or selected, it replaces the placeholder in the card area, as illustrated in Figure 13.2-4. The Classify button transitions to an active green state. A small camera icon overlaid at the bottom-right of the image card allows re-capture without leaving the screen. Tapping Classify uploads the image to the backend API via multipart POST. A loading indicator is shown during the network call (typically 1–2 seconds). On success, the user is navigated to the Classification Result screen.Classification Result Screen — Known Species

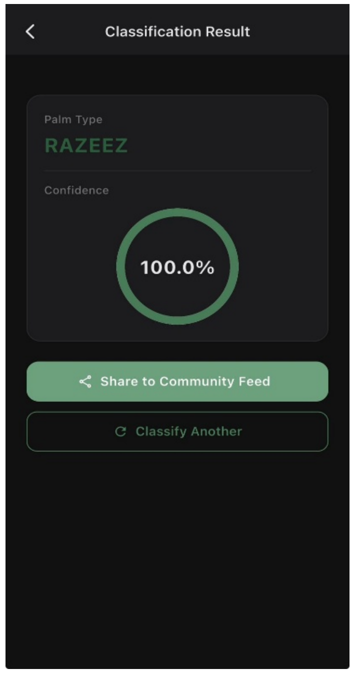

Figure 13.2-5 Classification Result Screen — Known Species

For a confidently identified species, this screen, shown in Figure 13.2-5, presents:

- **Palm Type:** The predicted class (KHALAS, RAZEEZ, or SHISHI) displayed in the brand green colour, with the uploaded frond image shown above it in a card.
- **Confidence:** A circular progress gauge showing the model's softmax confidence as a percentage (e.g., 100.0% for RAZEEZ). The gauge ring fills proportionally to the confidence value.

Two action buttons follow: "Share to Community Feed", opens a caption dialog and posts the result to the community; and "Classify Another", returns to the empty classification screen.

13.2.5 Classification Result Screen — Unknown Species

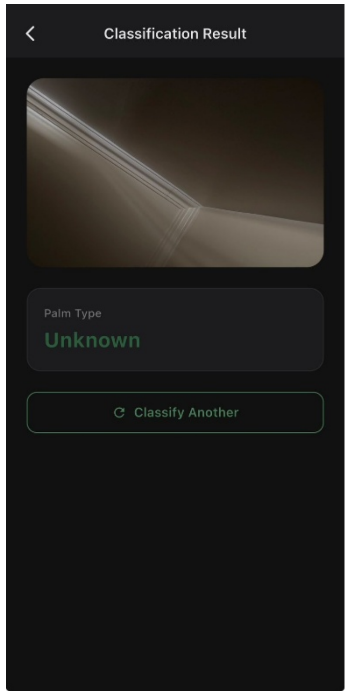

Figure 13.2-6 Classification Result Screen, Unknown Species

When the submitted image does not match any of the three known varieties with sufficient confidence, the model returns "Unknown," as shown in Figure 13.2-6. This state omits the confidence gauge and the Share button, displaying only "Palm Type: Unknown" and a "Classify Another" button. This graceful degradation prevents incorrect data from propagating into the community feed.

13.2.6 Community Feed Screen

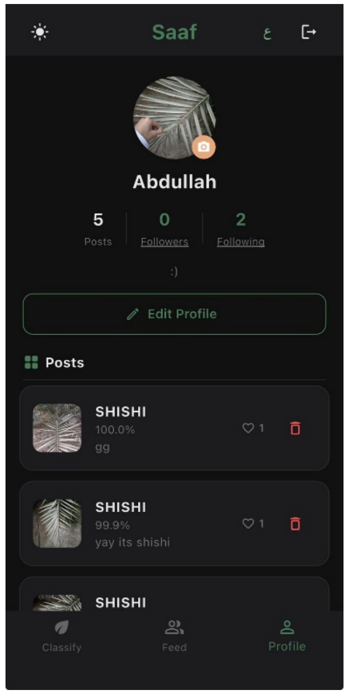

Figure 13.2-7 Community Feed Screen

As illustrated in Figure 13.2-7, the Feed tab displays a chronological stream of posts shared by all users. Each post card contains: user avatar, display name, and post date; the frond photograph; a classification badge (e.g., SHISHI · 99.9%) anchored below the image; an optional text caption; and an engagement row with a heart (like) icon and a comment-bubble icon with respective counts. Users who own a post see a red trash icon to delete it. The feed supports pull-to-refresh and paginated loading.

13.2.7 User Profile Screen

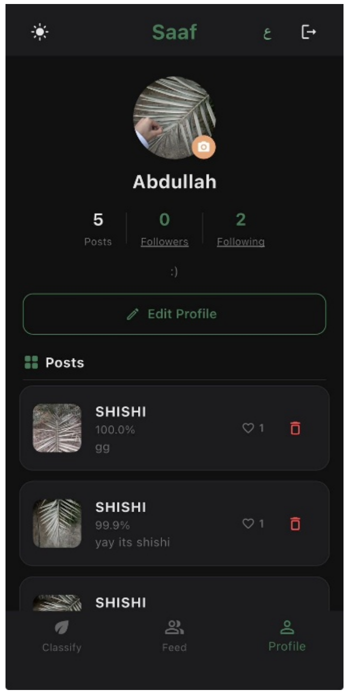

Figure 13.2-8 User Profile Screen

The Profile tab, shown in Figure 13.2-8, displays the authenticated user's personal space. A circular avatar (editable via a camera-badge tap) is accompanied by the display name, three stat counters (Posts, Followers, Following), and an optional bio. An "Edit Profile" button opens a form to update name, bio, and avatar. The lower portion lists all posts the user has shared, each showing a thumbnail, classification label, confidence score, caption, like count, and a delete button. Tapping the logout icon in the top-right corner of any main screen ends the session.

14 Final Testing Process and Results

This chapter consolidates the testing activities conducted across all project phases and presents the final validated results of both the AI classification model and the full application system.

14.1 Testing Methodology Summary

Testing ran at three levels: individual model experiments, REST API integration with the Flutter client, and end-to-end functional test cases. All experiments were designed around realistic field conditions, variable lighting, different camera angles, images taken during the day and at night.

14.2 AI Model - Final Consolidated Results

Seven experiments compared model architectures across two dataset configurations: daytime-only training versus combined day-and-night training. Table 14.2-1 shows the full picture.

Experiment	Model	Training Data	Test Data	Accuracy
Exp 1	ViT_b_16	Day Images Only	Night Images Only	**55.33%**
Exp 2	ViT_b_16	Combined (Day + Night)	Night & Day (10%)	**92.00%**
Exp 3	KNN	Combined (Day + Night)	Night & Day (10%)	**91.02%**
Exp 4	SVM	Combined (Day + Night)	Night & Day (10%)	**90.88%**
Exp 5	ConvNext_small	Combined (Day + Night)	Night & Day (10%)	**98.37%**
Exp 6	ConvNext_Tiny	Combined (Day + Night)	Night & Day (10%)	**97.55%**
Exp 7	ConvNeXt-Small (5-fold ensemble)	Combined (Day + Night)	Night & Day (10%)	**98.23%**

Table 14.2-1 Final Model Accuracy Summary

ConvNeXt-Small (Exp 5) had the highest raw accuracy at 98.37%. But both single-model runs, Exp 5 and Exp 6, shared the same failure: anything outside the three training classes got labeled Khalas. That's not a rounding error in the metrics. It's the kind of thing a farmer notices the first time they point a camera at something unfamiliar.

The ensemble (Exp 7) was built to fix this. Five ConvNeXt-Small models trained on different data splits, all running at inference time, predictions averaged. Accuracy came in at 98.23% - 0.14% below the single model - with more balanced recall across all three varieties and no consistent mislabeling of out-of-distribution inputs. That's the model in production.

14.2.1 Per-class performance - ConvNeXt-Small ensemble (Exp 7)

KHALAS	1.00	0.99	0.99	245 images
RAZEEZ	0.97	0.98	0.98	~248 images
SHISHI	0.98	0.97	0.98	~242 images
Weighted Avg.	0.98	0.98	0.98	~735 images

Table 14.2-2 Per-Class Performance Metrics for ConvNeXt_Small

14.2.2 Per-class performance metrics for the ConvNeXt-Small ensemble

All three classes came in at F1 0.98 or above across 735 test images. Khalas hit perfect precision at 1.00 notable given it was the class the single models over-predicted. The ensemble did not just maintain accuracy; it distributed it evenly.

15 Conclusion and Recommendations

15.1 Summary of Challenges and Solutions

The development of Saaf required navigating a set of interconnected technical and organizational challenges. Table 15.1-1 consolidates the five primary challenges and the solutions applied.

Challenge	Root Cause	Solution Applied	Outcome
Limited dataset availability	Palm frond images of the three species are not publicly available in large quantities	Direct field collection from a local farm; augmentation (flip, brightness jitter, rotation)	Sufficient training data; model generalized across lighting conditions
Day/night performance gap	Models trained only on daytime images failed on night images (ViT: 55.33%)	Combined day-and-night training set; ConvNeXt_Small selected for robustness	97.55% accuracy on mixed test set
Specialized social network design	Generic platforms lack agricultural context and AI integration	Custom Feed with embedded classification badges and domain-specific posting flow	Farmers can share verified species identifications directly from the app
AI-mobile integration complexity	Flutter client needed to upload images and receive structured predictions reliably	RESTful API (FastAPI) with multipart upload; JSON response with class and confidence	Sub-1.2 s end-to-end latency; all 16 functional test cases passed
User-friendliness for non-tech farmers	Target users may have limited smartphone experience	Minimal-step UI (3 taps to classification); Arabic/English toggle; large touch targets	Clean, accessible interface validated through user walkthroughs

Table 15.1-1 Challenge-Solution Summary

15.2 Reflections on Lessons Learned

The Saaf project yielded several non-obvious insights that are expected to be of value to future researchers and practitioners.

- **Data quality over quantity:** The most significant accuracy leap came not from collecting more images, but from ensuring the training set reflected the full distribution of deployment conditions, specifically by including nighttime images. This underscores the importance of understanding the deployment environment before finalizing the data collection strategy.
- **Architecture efficiency matters for deployment:** The 0.82% accuracy gap between ConvNeXt-Small (98.37%) and ConvNeXt-Tiny (97.55%) made the

Tiny look like the smarter pick at first. Lower memory, faster inference, cheaper to run on a compute-billed server, those are real advantages. But both single models had a problem that never showed up in the numbers: anything outside the three training classes got labeled Khalas. Every time. That kind of failure is invisible in a benchmark and obvious the moment a farmer points at something the model wasn't trained on.

The 5-fold ConvNeXt-Small ensemble fixed it. More compute than a single model, yes, but recall balanced out across all three varieties and the Khalas default on unfamiliar inputs went away. The lesson is pretty simple: accuracy alone doesn't tell you how a model fails. Memory, latency, and what happens on inputs the model has never seen, those matter too, especially when real people are relying on it.

- **Community features require equal design attention:** The social networking component required careful UX consideration to ensure farmers found it intuitive and trustworthy. Early informal tests revealed that users were uncertain about what "sharing" a classification meant. Redesigning the share flow to preview the post before confirmation eliminated this confusion.
- **Bilingual support requires early planning:** Implementing Arabic RTL layout after the English UI was largely complete required restructuring several widget trees. Future projects targeting Arabic-speaking audiences should adopt internationalization scaffolding from the first sprint.
- **Cross-functional coordination is critical:** The project required parallel progress across Deep learning, backend engineering, and mobile development. Coordination delays, particularly around agreeing on the API contract schema, caused short-term bottlenecks. Establishing a shared API specification document early would have streamlined integration considerably.

15.3 Suggestions for Further Work and Scalability

While the current version of Saaf fulfills all stated objectives, several directions for extension and scaling are identified:

- **Expand the classification scope:** The current model targets three palm varieties. Saudi Arabia is home to dozens of commercially significant date palm cultivars. Future work should extend the dataset to cover additional varieties, potentially in collaboration with the Ministry of Environment, Water and Agriculture or KFU agricultural research centers.
- **Disease and health detection:** The frond-image pipeline developed for species classification is directly applicable to palm disease detection. A parallel model trained to identify common palm diseases (e.g., Bayoud disease, red palm weevil damage) would substantially increase the agronomic value of the platform.
- **Crowdsourced dataset expansion:** Each classification shared to the community feed generates a labeled image. With appropriate user consent mechanisms, this data can be used to continuously retrain and improve the model, creating a virtuous cycle between user activity and model performance.
- **Scalable infrastructure:** The current single-VM deployment is adequate for a prototype. Production scaling would require containerized horizontal scaling (Kubernetes), a managed database service, and a CDN for image delivery. These changes are straightforward given that the application is already fully containerized.

15.4 Limitations

Despite the achievements of the Saaf system, several limitations should be acknowledged:

- The classification model is restricted to three palm varieties — Khalas, Sheeshi, and Rzeez — out of the many cultivars found across Saudi Arabia.

- The training dataset was collected from a single local farm, which may limit the model's ability to generalize to palm trees grown in different regions or environmental conditions.
- The model does not account for variations in frond appearance caused by palm tree age or disease conditions, both of which can significantly affect visual characteristics.
- The current infrastructure relies on a single virtual machine deployment, which is not yet scaled for production-level usage.

16 Appendices

16.1 User Manual

This manual provides step-by-step instructions for using the Saaf mobile application. It is intended for end users (farmers and agricultural workers) and assumes basic familiarity with a smartphone.

16.1.1 Getting Started

- **System Requirements:** Android 8.0 (API 26) or higher, OR iOS 14.0 or higher. A working internet connection is required for classification and community features.
- **Installation:** Install the Saaf APK (Android) or IPA (iOS) provided by the project team. On Android, enable "Install from unknown sources" in device security settings if prompted.

16.1.2 Creating an Account

- **Step 1:** Open the Saaf app. The Registration screen appears on first launch.
- **Step 2:** Enter your Full Name, Email, Password, and Confirm Password. The password must be at least 8 characters.
- **Step 3:** Tap Register. On success, you will be taken directly to the main Classify screen.
- **Step 4:** To switch the language to Arabic at any time, tap the globe icon (العربية) in the upper-right corner.

16.1.3 Logging In

- **Step 1:** Enter your registered Email and Password on the Login screen.
- **Step 2:** Tick "Remember Me" to remain logged in across sessions.
- **Step 3:** Tap Login. To reset a forgotten password, tap "Forgot Password?" and follow the email instructions.

16.1.4 Classifying a Palm Frond

- **Step 1:** Tap the Classify tab (leaf icon) in the bottom navigation bar.
- **Step 2:** Tap Camera to photograph the frond, or Gallery to select an existing photo. Position the camera 30–60 cm from the frond with good lighting.

- **Step 3:** Once the image appears in the card, tap the green Classify button.
- **Step 4:** Wait 1–2 seconds. The Classification Result screen will display the Palm Type (KHALAS, RAZEEZ, SHISHI, or Unknown) and confidence score.
- **Step 5:** Tap "Share to Community Feed" to post the result with an optional caption, or "Classify Another" to start over.

- **Note:** If the result shows "Unknown," the image may not be a clear palm frond photograph. Try again with a closer, well-lit frond image.

16.1.5 Using the Community Feed

- **Step 1:** Tap the Feed tab (people icon) to browse posts from other farmers.
- **Step 2:** Scroll through the feed. Each post shows the farmer's name, date, frond image, classification badge, and optional caption.
- **Step 3:** Tap the heart icon to like a post. Tap the comment icon to open the comment thread and add a reply.
- **Step 4:** To delete one of your own posts, tap the red trash icon in the top-right corner of the post card.

16.1.6 Managing Your Profile

- **Step 1:** Tap the Profile tab (person icon) in the bottom navigation bar.
- **Step 2:** Tap "Edit Profile" to update your display name, bio, or profile photo (tap the camera badge on the avatar).
- **Step 3:** Your complete post history is listed below the profile header with thumbnail, classification result, and engagement metrics.
- **Step 4:** To log out, tap the exit icon in the top-right corner of any main screen.

16.2 Technical Documentation

16.2.1 System Architecture Overview

- The Saaf system follows a three-tier architecture:
  - **Tier 1 - Presentation:** Flutter mobile application (Android/iOS). Handles all user interactions, image capture, and data display.

- **Tier 2 - Application:** Python FastAPI server. Exposes the classification REST endpoint; loads and runs the ConvNeXt_Small model; manages business logic.
- **Tier 3 - Data:** PostgreSQL database (user data, posts, social graph)

16.2.2 REST API Endpoints

Method	Endpoint	Description	Auth Required
POST	/classify	Upload a palm frond image; returns predicted class and confidence	Bearer token
POST	/auth/register	Create a new user account	None
POST	/auth/login	Authenticate and receive JWT token	None
GET	/feed	Retrieve paginated community posts	Bearer token
POST	/posts	Create a new post (share classification result)	Bearer token
DELETE	/posts/{id}	Delete a post owned by the authenticated user	Bearer token
POST	/posts/{id}/like	Toggle like on a post	Bearer token
POST	/posts/{id}/comments	Add a comment to a post	Bearer token
GET	/users/{id}/profile	Retrieve user profile and post history	Bearer token
PATCH	/users/{id}/profile	Update profile name, bio, or avatar	Bearer token

Table 16.2-1 REST API Endpoint Reference

16.2.3 Classification Request and Response Schema

- **Request:** HTTP POST to /classify with Content-Type: multipart/form-data. Field name: "file". Accepted MIME types: image/jpeg, image/png. Maximum size: 10 MB (images are compressed client-side before upload).
- **Successful Response (200 OK):** { "predicted_class": "RAZEEZ", "confidence": 1.00, "all_probabilities": { "KHALAS": 0.00, "RAZEEZ": 1.00, "SHISHI": 0.00 }}
- **Unknown Response (**200 **OK):** { "predicted_class": "Unknown", "confidence": null, "all_probabilities": { "KHALAS": 0.34, "RAZEEZ": 0.33, "SHISHI": 0.33}}

16.2.4 Database Schema (Core Tables)

Table	Key Columns	Description
users	id (UUID), email, display_name, bio, avatar_url, created_at	Registered user accounts
posts	id, user_id (FK), image_url, palm_type, confidence, caption, created_at	Community feed posts
likes	user_id, post_id, created_at	Many-to-many like relationships
comments	id, post_id (FK), user_id (FK), body, created_at	Post comment threads
follows	follower_id, following_id, created_at	Follower/following social graph

Table 16.2-2 Core Database Schema

16.2.5 Deployment Configuration

Component	Technology	Version	Notes
Mobile App	Flutter / Dart	3.x / ^3.0	Release build; debug logging disabled
API Server	FastAPI + Uvicorn	0.104 / 0.24	Gunicorn with 2 workers; port 8000
AI Framework	PyTorch + torchvision	2.1 / 0.16	CPU inference; CUDA optional
Database	PostgreSQL	15	Hosted on same VM; pg_dump daily backup
Reverse Proxy	Nginx	1.24	HTTPS via Let's Encrypt; HTTP redirect
Containerization	Docker + Compose	24.x / 2.x	Single compose file for API and DB
OS / Hosting	Ubuntu 22.04 LTS	22.04	2 vCPU, 4 GB RAM cloud VM

Table 16.2-3 Deployment Configuration Reference

17 Work Timeline

17.1 High-Level Timeline / Gantt Chart

Figure 17.1-1 presents the Gantt chart for the full project timeline, covering both the proposal and implementation phases from September 2025 through May 2026, including all milestones and defense submissions.

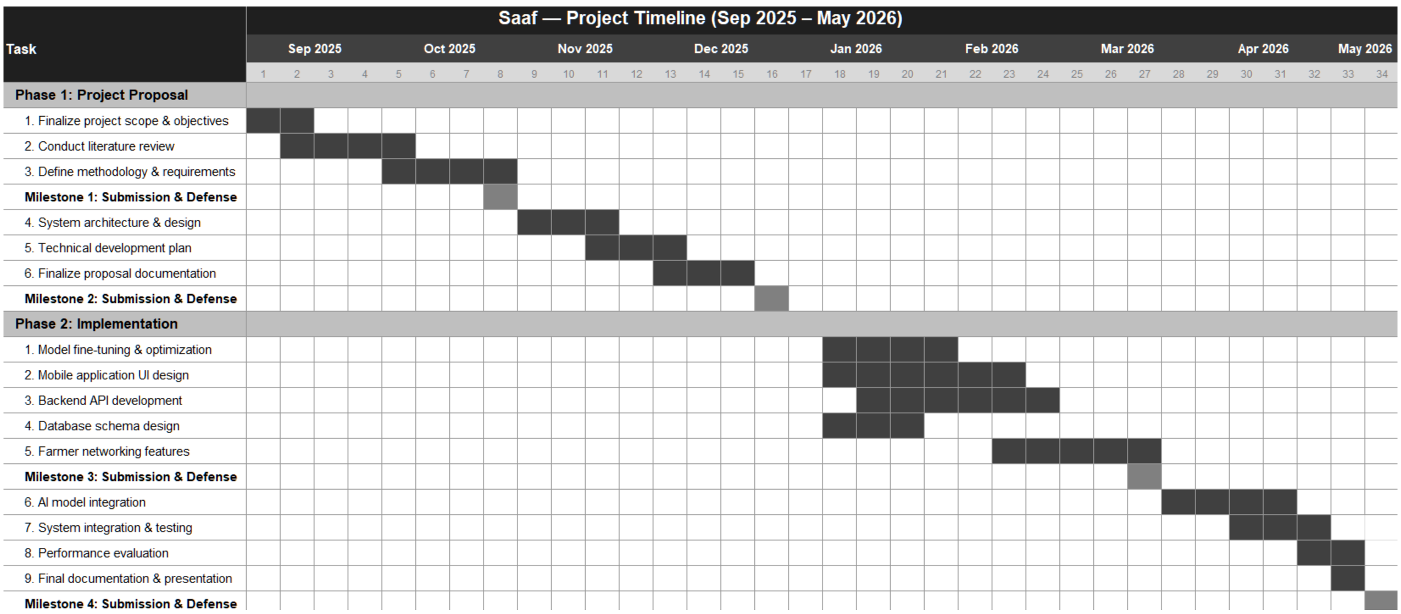

Figure 17.1-1 Graduation Project Timeline

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